



Securing water resources, safeguarding our future

Water Resources Performance Report



Conservation of Lamu Sand dune by WRA

**A report to the public from Water Resources Authority
for the Period Year 2022 -2023**

Chairman Remarks

The Water Resources Authority serves the public through implementation of its mandate of regulating the management and use of water resources in Kenya. It undertakes this as anchored in the Water Act 2016, Water Resources Regulation 2021, National Water Resources Strategy 2020-2025 and Water Policy 2021. During the period 2022-2023, the Authority was able to make commendable progress in various thematic areas.



There was remarkable performance and increase in the number of Water Abstraction and Pollution surveys conducted, Sub catchment plans developed and implemented by Water Resources Users Associations (WRUAs), Strengthened data management, Protection of water catchment areas, Rehabilitation of water monitoring stations, and Riparian marking and pegging.

The Authority re-engineered the following processes; Revenue management by converting paper-based receipts to automated system receipts, Meter reading by adopting mobile Application which helps in capturing the meter reading and transmitting data into the water meter data management system real-time. Billing by dispatching payment invoices to customers through SMS and e-mail.

The increasing competing needs for water resources threatens its sustainability and livelihoods. The dire state of water quality in the country calls for efforts to be made by all stakeholders to ensure achievements gained are evidenced at grassroots levels among the citizens.

Catchment protection especially in the water towers needs to be prioritized to support livelihoods and ensure water flows to the taps. This cannot be achieved without increase in funding from the government and external partners. We call upon all our strategic stakeholders to collaborate with us in the implementation

of our mandate to be able to serve all Kenyans as we fulfill their constitutional rights of access to clean and safe water.



Donald K. Murgor
Chairman Management Board

CEO Remarks

The Water Resources Authority serves the public through implementation of its mandate of regulating the management and use of water resources in Kenya. In facilitating efficiency and effectiveness of its services, the Authority continuously automates its processes to fulfil constitutional right of Citizens of access to clean and safe water.



During the Period 2022-2023, the public was able to submit permit application through E-permitting and access permit processing services in the E citizen portal. Additionally, more smart meters were procured and installed by the water abstractors, thus improving data collection for informed decision making.

This annual Performance Report communicates to the public status of water resources issues across the six basin areas in Kenya that covers the forty-seven county governments. The main issues reported are on Water Allocation Criteria, Engagement of Communities through WRUAs in water resources, water quality , flood mitigation, Water Resources Assessment and Monitoring, Enforcement, Economic and Financial Management, Information management, Complaints resolution and WRA Research Publications.

The Authority achieved an excellent score (2.0473) in its Performance Contract, placing it at position 5 out of 394 public institutions in Kenya. This illustrates its commitment to efficient and satisfactory service to the citizens.

Mohamed M. Shurie, OGW, MBS
Chief Executive Officer

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Executive Summary

During the Financial Year 2022/2023 the Authority implemented its mandate of regulating the management and use of water resources in Kenya. The Authority eased access to its vital service of water allocation through enabling online permit application, and access through e citizens. In improving real time data acquisition for decision making (flood and drought early warning) water resources availability. 120 stations have been upgraded to telemetry status out of an optimum 500 monitoring stations countrywide for climate monitoring, 377 out of 500 stations will be equipped with additional sensors for water quality monitoring. Groundwater development has increased therefore, the Authority continues to equip dedicated monitoring boreholes, with data loggers. Exploratory boreholes are drilled in areas which aquifers are not well known to inform the characteristics/properties of the aquifer. In the next 5 years, 80 boreholes will be drilled and equipped for telemetric data collection and transmission for improved groundwater resources management. The Authority will also focus on implementation of nature-based solutions undertaken by Water Resource Users Associations (WRUAs). In flood mitigation the Authority develops Flood Early Warning System (FEWS) to forecast the likely flood events and provide alerts on the required actions (evacuation, caution, relocation) to both the residents and disaster management organizations/agencies for pre-emptive action. The FEWS have an input from the Kenya Meteorological Department (KMD) rainfall forecasts as rainfall is a key input to the system. In engaging communities through Water Resource Users Association (WRUAs) WRA has mapped and gazette 6 basin areas covering the entire country with a delineated optimal WRUA boundary of 1237 nationally. During the reporting period cumulatively 760 WRUAs were established, 16 Sub Catchment Management Plans (SCMP) developed, 22 SCMPs reviewed and implementation undertaken. The Authority enhanced its staff competencies, professional training in Change management training, Information Communication Technology (ICT), Integrated Water Resource Management (IWRM) and Quality Management Systems laboratory certification. The staff have up to date registration with professional memberships listings in various professions that enhances competency among staff and increases efficient and effective service delivery to the public.

1 Introduction

1.1 Content of the report

The report contains the Authority performance in the year 2022/2023 in its various Basin Areas implementing its mandate of regulating the management and use of water resources.

1.2 River Basin Water Resources Management

WRA has a national coverage area and manages its water resources along river basins, this requires that a particular river be managed as a single unit from its source to the end. Kenya is naturally divided into 5 Drainage Areas or Basins, deriving their names from the dominate water body, namely:

- I. Lake Victoria Basin (divided into Lake Victoria North and Lake Victoria South for administrative purposes)
- II. Rift Valley Internal Drainage
- III. Athi Basin
- IV. Tana Basin
- V. Ewaso Ng'iro North Basin

1.2.1 Kenya Basin Offices

WRA has 3 operational (functional) levels with 33 offices comprising of:

- 1 National Headquarters located in Nairobi (NHIF Building);
- 6 Basin Area Offices;
- 26 Basin Area offices.

Table 1: WRA Sub Basin Areas within Basin Areas

<i>Basin Area Office</i>	<i>Sub Basin Area Office</i>
Lake Victoria North	Kitale, Eldoret, Siaya
Lake Victoria South	Kericho, Kisii, Kisumu
Rift Valley	Narok, Naivasha, Kabarnet, Kapenguria, Lodwar
Athi	Nairobi, Kiambu, Kibwezi, Loitoktok, Mombasa
Tana	Muranga, Kerugoya, Meru, Kitui, Garissa
Ewaso Ngiro North	Nanyuki, Rumuruti, Isiolo, Marsabit, Wajir

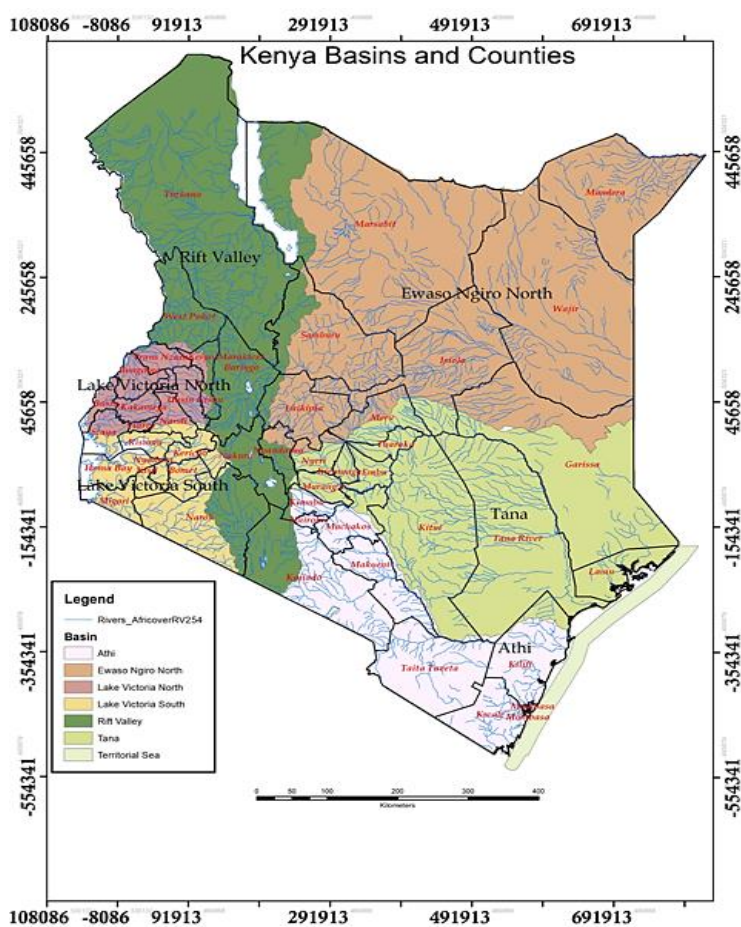


Figure 1: Map of WRA Basin Areas & Counties

1.3 Highlights of major achievement

- The Authority during FY 22/23 in its Performance Contract got an excellent score of 2.0473 which was a steady performance of the previous year the Authority had an excellent score of 2.0515 and was positioned 5 out of 394 institutions.
- Section 23 of the Act, enables the Authority to make a declaration in the Kenya Gazette, designating certain areas to be Groundwater Conservation Areas in order to take special measures to conserve groundwater resources. In the implementation of these provisions of the law, the Authority developed and facilitated the publication of Legal Notices for the protection of catchment areas and declaration of groundwater conservation areas. The Legal Notices together with the accompanying Management Plans and Explanatory Memoranda were in respect to the following Protected Catchment Areas and Groundwater Conservation Area Orders respectively:
 - i. The *Kabeere* Springs Catchment Protection Area Order, 2022;
 - ii. The *Kajulu* Hills Catchment Protection Area Order, 2022;
 - iii. The *Manguo* Springs Groundwater Conservation Area Order, 2022; and
 - iv. The *Marura* (Ewaso Narok) Swamp Catchment Protection Area Order, 2022

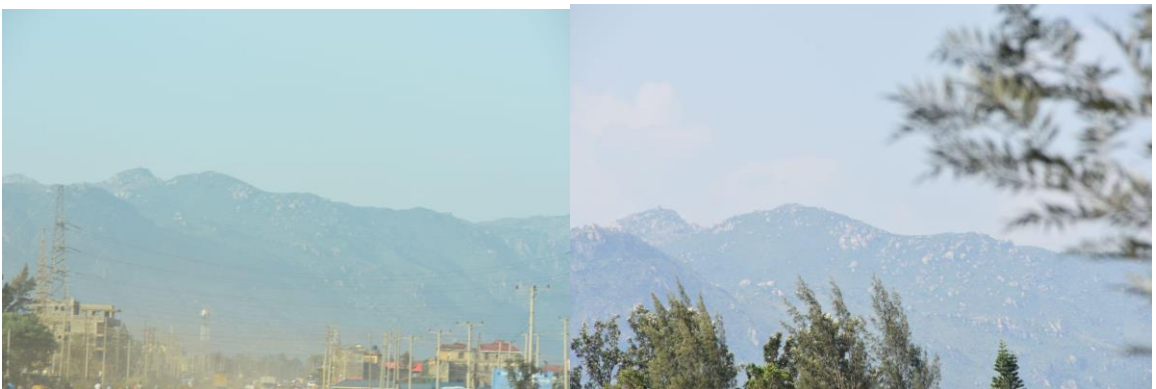


Figure 2: Gazetted Kajulu Hills

- The foregoing is an addition to the Catchment Protection Orders and Groundwater Conservation Area Orders already under implementation published through various Legal Notices in preceding years. The

Authority also commenced implementation of the National Water Policy, 2021 and the newly enacted Water Resources Regulations, 2021. The provisions in respect to water use fees were realized, albeit for a very short period of time before an injunction was issued by court, temporarily suspending the implementation. Separately, a similar case filed to challenge the implementation of the said Regulations was dismissed in favor of the government and government agencies, including WRA.



Figure 3: Gazetted Manguo Swamp

- Groundwater monitoring stations were increased by establishing 9 new monitoring stations, 7 Lamu, 2 Nairobi, for acquiring Groundwater data for decision making. Construction of Djabia in Mpeketoni used to harness rainfall that boosts water availability supporting the community to build resilience to climate change.
- The Authority was audited in readiness for ISO 9001:2015 Quality Management System re certification for WRA operations and functional levels



Figure 4: Conserved catchment

- The Authority forged partnerships and linkages with the aim of expanding and sustaining a diverse funding base for water resources. During this reporting period, developed 19No. surpassing the target of 13No proposals and enhanced partnerships that have been instrumental in supporting the institution's project implementation efforts.

The Authority strives to maintain its high achievement and work towards putting more efforts, resources and seeking partnerships in areas that need improvement. As the automation of services progresses well, service delivery to the citizens will tremendously improve ensuring customer satisfaction and giving services to the public timely

2. Regulating the management and use of water resources

The Water Resources Authority implements Water Resources Strategy 2020-2025 and Water Resources Regulations 2021, this provides programmes for the regulation, monitoring, protection, conservation and management of the Water Resource. Water resources management intervention measures undertaken to improve water availability and reduction of adverse effects on water resources include the following among others: flood mitigation,

catchment areas restoration; riparian zone protection and conservation; control of soil erosion; construction of sand dams and earth dams; protection of wetlands and springs, water harvesting and safe storage, groundwater exploration, installation and rehabilitation of water resources monitoring networks.

Development and implementation of integrated water resources plans and strategies, development and implementation of Water Allocation Plans, pollution monitoring and control; Water Resource Users Associations programmes on water resources management, exploring the innovative solutions for waste management.

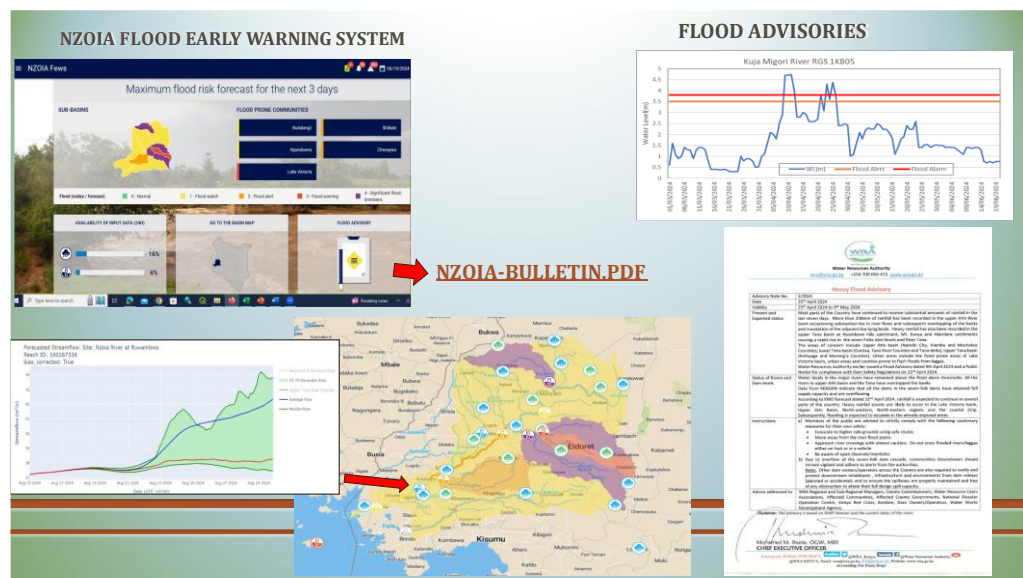


Figure 5: Flood Mitigation through Flood Advisory

2.1 Per Capita Water Availability

All the available water worldwide is contained in what is referred to as the Hydrological cycle and which defines what is the Water per capita (amount of water available for each person per year). The water per capita is area specific and depends on the water availability in an area. With the growing population, Kenya faces enormous challenges in the management of its limited water resources across most sectors of the economy, making water resources management a high priority. The globally acceptable per capita water

availability is 1,000m³. However, in Kenya the average per capita is around 620m³ and is continually decreasing. Increase of this figure is possible through the conservation and protection of water resources and catchment areas.

2.2 Water Demand Vs Availability

According to the National Water Master Plan 2030, the total water demand will increase significantly against the available water by the year 2030. In the year 2010, the total demand was 3,218MCM/year against available 22,564 MCM/year (surface and groundwater yields). The demand will rise to 21,468 MCM/year in the year 2030 against available 26,634 MCM/year as shown in the table below (*Source: National Water Master Plan (NWMP 2030)*)

Table 2: Available Water Against Demand per Basin Area

<i>Basin Area</i>	<i>Area (Sq Km)</i>	<i>2010</i>		<i>2030</i>		<i>2050</i>	
		<i>Demand</i>	<i>Available</i>	<i>Demand</i>	<i>Available</i>	<i>Demand</i>	<i>Available</i>
Lake Victoria North	18,374	228	4,742	1,337	5,077	1,573	5,595
Lake Victoria South	31,734	385	4,976	2,953	5,937	3,251	7,195
Rift Valley	130,452	357	2,559	1,494	3,147	1,689	3,903
Athi	58,639	1,145	1,503	4,586	1,634	5,202	2,043
Tana	126,026	891	6,533	8,241	7,828	8,476	7,891
Ewaso Ng'iro North	210,226	212	2,251	2,857	3,011	2,950	1,810
Total	575,451	3,218	22,564	21,468	26,634	23,141	28,437

Source National Water Master Plan (NWMP 2030)

Some Basin Areas (Athi, Tana and Ewaso Ngiro North) will suffer water resource deficit as the demand will be higher than availability by the year 2030. However, the deficit might be realized earlier due to the impacts of climate change and if no proactive water resources and catchment management activities are undertaken (catchment degradation, pollution, riparian reserve and catchment areas encroachment).

2.3 Advanced Metering Infrastructure

WRA continues to implement Advanced Metering Infrastructure (AMI) for Online Capture of Water Meter Data System. In this financial year the Authority rolled out its permit application on E Citizen and the public can now access permit services online.

Water Resource Information System

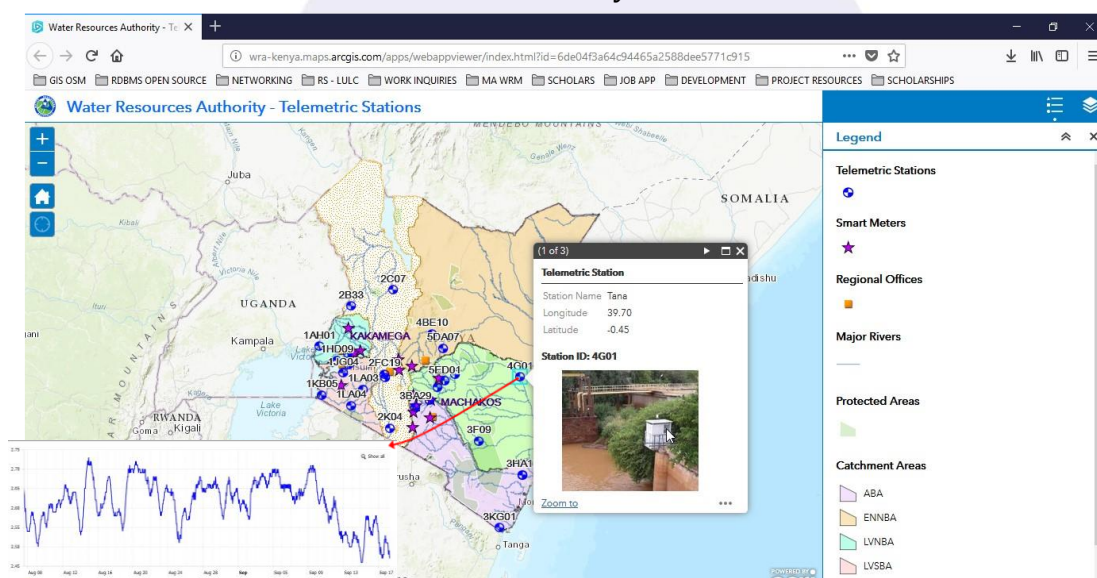


Figure 6: Water Resources Information System

2.4 Catchment Protection

The Water Act 2016 mandates the Authority to protect catchment (Section 22) and groundwater conservation areas (Section 23) by preparing a Management Guidelines/Plan for the Protected (surface water) or Groundwater Conservation Area (groundwater). Prepared in conjunction with the stakeholders within the targeted area. The Management Guidelines describe the area and water resources issues, proposed interventions required to protect and conserve the

water resources and proscribed activities within the protected or groundwater conservation area. The Guidelines are published as part of the Order in the Kenya Gazette, imposing such requirements or prohibiting some activities as may be considered necessary for the conservation of water resources.

Where deemed necessary, tilling or cultivation, clearing of indigenous trees or vegetation, building of permanent structures, disposing of waste, soil excavation or quarrying, and growing of exotic species that may have adverse effect to the water resource, amongst others may be expressly proscribed.



Figure 7: Catchment Restoration Prescribed activity of tree growing

During the period under review, WRA initiated the gazettement process for four vulnerable catchments - Lari and Karai Swamps in Kiambu County, Lake Ol'bolosat in Nyandarua County and Mwangea Hills in Kilifi County. The Management Guidelines and gazettement orders were developed for the four catchments and have since been submitted to the Office of the Attorney General for review and subsequent transmission to Parliament for consideration and approval. Currently, the process is at an advanced stage and is expected to be finalized soon.



Figure 8: Lake Olobollosat

2.5 Upgrading of countrywide Hydro met stations

The Authority has been upgrading river gauging stations to telemetric to record and transmit real time data. For proper water resources management this improves reliability and timeliness of the data and provides adequate infrastructure for hydrological observation in the country. The economic and social benefits of the upgraded stations include efficiently managed water resources with sufficient knowledge and enable floods and drought to be predicted thereby providing adequate mitigation measures to be taken. One manual surface water monitoring station in Nairobi River RGS 3BA29 was upgraded to telemetric and configured to transmit data to the WRA server.



Figure 9: Telemetric Station along Nanyuki River

2.6 Resource Mobilization and Partnership

The Authority undertakes resource mobilization and partnership, one of the ways is through proposal writing that is submitted to Treasury through the Ministry of Water Sanitation and Irrigation. During the period under review 19No. proposals were developed. The proposals focused on a range of strategies for water resource regulation, management and climate resilience in Kenya. These proposals are aimed to address water scarcity, improve water quality, and support community resilience to climate change impacts. They emphasize the importance of ecosystem conservation, community involvement, and technological advancements in achieving sustainable water management and climate resilience.

Partnerships play a crucial role in supporting the WRA to achieve its mandate. They enable the pooling of resources, sharing of expertise, and fostering of innovation. Collaborations with various stakeholders, including government bodies, private entities, and development partners, have led to significant advancements in water resource management. The Ministry of Water & Sanitation and Irrigation support the institution in linking the institution to partners through the proposals developed. The key ongoing Partnerships and Projects are:

- Horn of Africa Groundwater for Resilience Project (HoAGW4R): A World Bank-funded initiative aimed at enhancing groundwater resilience.
- Blue Deal Programme: Supported by the Dutch Water Authority, focusing on water resource management in Tana Basin.
- Catchment To Tap (C2T) Project: A partnership with World Wildlife Fund (WWF) to implement sustainable water management practices.
- Japan International Cooperation Agency (JICA): Capacity development in surface water monitoring and management in Tana Basin.
- Enhancing Community Resilience and Water Security: A Green Climate Fund (GCF) funded project for climate change adaptation and water security which has been approved by GCF awaiting implementation.
- Towards Ending Drought Emergencies (TEWNDE): Ecosystem Based Adaptation in Kenya's Arid and Semi-Arid Rangelands- Aimed at ending drought emergencies through ecosystem-based adaptation.
- Kenya Water Security Climate & Resilience Project: undertaking watershed management activities supported by the World Bank.

Partnerships support WRA in fulfilling its mandate by bringing together diverse expertise and resources, leading to improved water security, regulation, management, and sustainability.

2.6.1 Kenya Water Security and Climate Resilient Project

During the FY 2022/2023, the Authority through the Kenya Water Security and Climate Resilience Project (KWSCR) funded by World Bank supported 5 WRUAs in the Nzoia basin to implement watershed conservation activities. These activities were aimed at reducing sediments in the Nzoia River basin which is expected to reduce floods downstream and also prolong the lifespan of the dykes and irrigation infrastructure downstream. The WRUAs undertook the following activities.

- Bukhungu WRUA conserved 120Ha of land through marking and pegging 20Km of Bukhungu River riparian, protecting 1 No spring, Installation of 3No.10M3 Rain water harvesting tanks, installation of 20 No. bee hives as a livelihood activity, Dairy feeds production – feed

Pulveriser and store construction, Daily milk processing and marketing – processing equipment and automated dispenser and local poultry eggs production, incubator and marketing.

- Mumias WRUA conserved 120 ha of land through marking and Pegging 20Km of Mumias River riparian, protection of 1 no. spring, terracing 10Km of farm land to conserve 35Ha of land, establishment of Vetiver bulking site 1No., Installation of 10M3 Rain water harvesting tanks 3No., Dairy cows for milk production 11No., training farmers good agricultural practices by undertaking demonstration farm through planting high value vegetables in 1 acre land, production of Assorted seedlings production 0.25acre and Improved poultry production 600No. as a livelihood activity.
- Isiukhu WRUA conserved 120Ha of land through marking and Pegging 20Km of Isiukhu River riparian, protection of 2 No. springs, Terracing 15Km to conserve 52.5Ha, Gabion construction 12No. to conserve 24Ha along Isiukhu River, Installation of 10M3 Rain water harvesting tanks 2No., installation of 30 No. bee hives Dairy cows for milk production 15No. and Dairy goats for milk production 22No. as as livelihood activity for income generation.
- Safu conserved 48Ha of land through marking and Pegging 8Km of Safu River riparian, protection of 3 No. springs, terracing 10Km farm land to conserve 35Ha of land, installation of 10M3 Rain water harvesting tanks 5No., promotion of Fish farming 3No. ponds (3000 fingerlings) and poultry production 900 chicks and promotion of piggery 1 pig house with 6No pigs as livelihood activity for income generation.
- Kisama WRUA conserved 72 ha of land through marking and pegging 12km of Kisama River riparian, protection of 3 no. Springs, terracing 12Km of farm land to conserve 42Ha, Installation of 10M3 Rain water harvesting tanks 3No., Seedlings production -tree nursery establishment 1No (5000 assorted tree seedlings and 2000 bamboo), Promotion of Fish farming 5No. ponds (7500 fingerlings), Poultry production 850 No. chicks, Fodder production- Bracharia grass farming 2 Acres & 1No. Store and bee keeping – 50No beehives in 2No.(bee houses) for income generation.

In completion of activities of the Nzoia watershed, the Authority plans to undertake activities on watershed management in Mwache catchment area in the new financial year.

2.6.2 Horn of Africa Groundwater for Resilience Project (HoAGW4RP)

The HoAGW4R Project aims to increase sustainable access and management of groundwater resources in the borders of the Horn of Africa. Water Resources Authority are obligated to carry out five (5) activities namely;

- Mobilization, establishment, and capacity building of WRUAs covering the selected NEDI aquifers areas within the aquifers in NEDI
- Preparation and Implementation of sub-catchment Management Plans (SCMPs) by WRUAs,
- Protection of aquifer recharge zones through gazettelement,
- Aquifer assessments, development and management plans for the 8 NEDI aquifers and aquifer management plans for Nairobi and Nakuru aquifers
- Establishing a monitoring network in the selected aquifers of NEDI and linking these to the upgraded groundwater management information system, including creating a mobile App to boost user interface and reporting on boreholes.
- Under this project Eight aquifer assessments and development of Management and development plans were in the process of procurement. The sub catchment areas will be identified for MAR and livelihood activities, negotiations on how to utilize the shared water resources between countries and resource mobilization to develop groundwater in the Horn of Africa region will be undertaken. In 2022/2023 the TOR for finalization of transboundary water resources policy and development of draft bill was undertaken and ready for engaging consultant



Figure 10: Water pan, infiltration gallery as a nature-based solution

2.6.3 Thwake Multipurpose dam catchment conservation activities

The Authority received support from the MoWSI under the Thwake Dam multipurpose project to undertake conservation of the Thwake catchment. Investment in conservation activities for the dam catchment is critical in ensuring sediment reduction flowing into the dam hence prolonging the lifespan of the dam and reduce costs for dredging the dam as a result of siltation. A study was undertaken and several hotspot areas were identified where conservation should be prioritized. In FY 2022-2023, three WRUAs were identified and funded to implement the conservation activities as reported below;

- Mbimbini WRUA managed to trench a total of 7,500 meters of terraces. The measurement was 1,364 meters 3ft wide x 2 ft wide and 6136 meters 2ft wide x2ft wide. They also planted 1,000 tree seedlings and 14 Kgs of grass seedlings. They have two demonstration farms where some terracing was done and tree seedlings and grass planted. The rest of terraces were distributed on the WRUA members and other community members' farms that had been identified to be severely degraded and signed MOUs with the WRUA to that effect. The WRUA also undertook

gabion construction and a total of 25 gabion boxes were laid on five sites within Makueni Sub County in Wote/ Nziu ward and Muva/ Kikumini ward at Kyamusoi village for Kyamusoi gabions, and also in Kaiti sub county, Ukia ward, Nzuuni village, Mutambukoni area for Kavutini site. The gabion walls were properly anchored to the sides of the gulleys. The contractor used the excavated soil used to backfill on back of gabion.

- Thwake Kalawa WRUA trenched a total of 9,376 meters of terraces measuring 3ft wide x 2 ft wide exceeding the targeted trenching meters by 1,376 meters. They also dug a retention ditch measuring 5x3 at a length of 236 meters as advised by the Makueni Sub County Agricultural Officer. The retention ditch was meant to catch and retain run off in the higher contours to reduce the velocity downhill so as to maximize seepage into the ground. The WRUA also undertook gabion construction in 4 different sites at Mbooni East, Kathulumbi Location at Syotuvali sub location. The gabion walls were properly anchored and the rocks reinforced the gabions into place. The contractor used the excavated soil used to backfill on the back of gabion. A total of 74 no. gabion boxes were constructed.
- Mukio Thwake WRUA trenched a total of 8,246 meters of terraces all measuring 3ft wide x 2 ft wide exceeding the target by 246 meters. They also managed to plant a total of 1,280 tree seedlings and 20 kilograms of grass. Gabion construction is still ongoing.

2.7 WRA Certification 9001:2015

The Authority is ISO 9001:2015 Certified, during this period the Authority undertook Internal and external quality audits in preparation for recertification

by Kenya Bureau of Standards (KEBS) for all WRA operations and functional levels.

2.8 Laboratory Operations

The Water Resources Authority (WRA) prioritizes clean and safe water for Kenyans nationwide. This is achieved through strategically placed network of laboratories offering comprehensive water quality and pollution testing. These laboratories are located in Kisumu, Kakamega, Nakuru, Naivasha, Murang'a, Embu, Nyeri (This serves the Ewaso Ng'iro North Basin Area), and Nairobi's Central Water Testing Laboratories (CWTL) (acting as the referral laboratory for WRA), showcase WRA's commitment to extensive water quality monitoring.

Streamlining Efficiency with LIMS: WRA's laboratories leverage state-of-the-art Laboratory Information Management Systems (LIMS) to streamline sample processing, data analysis, and reporting. LIMS ensures efficient sample tracking, minimizes errors, and facilitates data accessibility for informed decision-making.

Commitment to Accuracy: Advanced Instrumentation and Accreditation: In the last financial year (2022-23), these laboratories processed a significant number of samples – a staggering 6,000. Furthermore, their recent acquisition of an Atomic Absorption Spectrophotometer (AAS) for the CWTL in Nairobi exemplifies their ongoing pursuit of improved analysis accuracy



WRA's dedication extends beyond technology. WRA is actively pursuing ISO 17025:2017 accreditation for all their laboratories. This internationally recognized standard ensures adherence to rigorous quality control procedures, leading to greater confidence in the accuracy and reliability of test results for both internal and external clients. This accreditation also signifies WRA's commitment to international best practices, fostering collaboration with other accredited labs and facilitating global data exchange.

Comprehensive Water Quality Testing: The laboratory offers comprehensive analysis, encompassing chemical, microbial, and effluent testing. This covers approximately 45 parameters, ensuring a thorough assessment of water quality for various uses, including drinking water, irrigation, industrial activities, and wastewater treatment.

By continually investing in advanced technology like Laboratory Information Management System (LIMS) and Atomic Absorption Spectrometer (ASS), pursuing international accreditation, and training their staff, WRA demonstrates their unwavering commitment to providing Kenyans with reliable and accurate water quality data. This, in turn, safeguards public health and promotes sustainable water resource management.

2.9 WRA Capital Project

The Authority contributes to the vision 2030 by implementing capital projects which are strategically located in specific drainage basins that are vital in achieving the vision 2030. During the reporting period the Authority implemented the following capital projects:

2.9.1 Lamu Groundwater Conservation

The Authority has been carrying out Lamu Groundwater Conservation to protect and secure sand dunes aquifer for sustainable supply of fresh water from the sand dunes in Lamu Island which is threatened by seawater intrusion and encroachment since 2010. The project outcome is to ensure that the groundwater resources are secured and the recharge area completely protected

from encroachment. During the financial year, the Authority carried out the following: (i) Training on groundwater modeling done for 30 No. officers, (ii) Rain gauge installed in the compound of Lamu Water and Sanitation Company, over 10,000 trees planted, management committee for AMU Groundwater Conservation Area constituted and Water Allocation Plan developed and presented to stakeholders.



Figure 12: Lamu sand dunes walk board WRUA eco-tourism

2.9.2 Kikuyu Groundwater Conservation

This project is meant to protect and secure Kikuyu springs groundwater recharge through; Protection of key recharge wetlands in groundwater conservation area Delineation, survey and mapping of Karai and Lari swamp areas, Development and implementation of management plans guidelines. During the financial year, the Authority managed to carry out the following: (i) Karai and Lari swamps management guidelines developed and forwarded to the Ministry of Water, Sanitation and irrigation for gazzetement., (ii) Riparian marking and pegging for Karai and Lari swamps undertaken, and, (iii) Rainfall station established in Murengeti Primary School; Management Committee of the Kikuyu Springs Aquifer Groundwater Conservation Area formed; 2 tanks installed in Lusigetti Primary School and 3000 trees.



Figure 13: Groundwater monitoring station at Manguo Swamp

2.9.3 Implementation of Sub Catchment Management Plans

During the period under review (2022/2023), the Authority committed to undertake implementation of Sub-catchment management Plans (SCMPs) through its performance contract for the year under review as follows.

- Operationalize and capacity build the Kenya National Association of Water Resources Users Associations (KeNAWRUA)
- Form 2 WRUAs
- Undertake the construction of 1 sand dam
- Develop 2 SCMPs
- Install 5 No. Roof water Harvesting tanks

The following achievements were realized;

- The KeNAWRUA was registered by the Attorney General's office, and its elected officials were inducted on 5th-6th October, 2022 in Lukenya, Machakos County by the Authority and the WWF office. A physical office was allocated to KeNAWRUA WRA industrial Area and operational furnished with office items at the WRA Nairobi Sub-Basin Area in Industrial area and 3 No. Staff designated to the office.

- 3no. WRUAs Established namely; Pate WRUA –Pate Island, Lamu County; Tangelbei WRUA – Tiati, Baringo County and; Kangalema WRUA – Turkana County.
- 3 No. Sand dams constructed namely; Lesoit Sand dam – by Ilmunkush WRUA, Kajiado county; Kwa-Mweu-Kalimenza sand dam by Kalundu WRUA in Kitui County and Kalimangovu sand dam by Kalundu WRUA in Kitui County.
- 14 No. SCMPs developed namely for the WRUAs namely: Tangelbei, Pate, Upper Ikoo, Kuywa, Upper Kuywa, Nambale, Ikonzo, Ndokoko Lukakani, Mang’ombe Shariani, Birikani, Kirongwe, Lower Mwitasyano, Kaladoi and Nandigwa.
- 17 No. Roof water harvesting tanks were installed including;
 - a. 2 no. tanks installed by Bukhungu WRUA in Musingu and Ivono Primary schools
 - b. 2 no. tanks installed by Isiukhu WRUA in Itenyi and Muleche primary schools
 - c. 3 No. tanks installed by Mumias WRUA in Kashia Health centre, Shibinga and Ebwaliro primary schools
 - d. 5 No. tanks installed by Safu WRUA in ACK St. Lukes Segal school, Omalo, Nyamwanga, Siwar and Lugose primary schools
 - e. 7No Tanks installed by Ilmunkush WRUA in *Oloiborr’ajjik Primary School, Ormeleleki Primary School, Naretoi Primary School, Embuya Primary School, Elang’ata Nanyokie Primary School, Oldupai Primary School, Osarai Primary School*

Note: Implementation of the SCMP activities by the WRUAs was done through financial support from the Authority and additional support from development partners and stakeholders.



Figure 14: WRUA meeting

2.9.4 Drilling of Exploratory Borehole

This project entails drilling and equipping of dedicated boreholes in strategic aquifers, monitoring assessment and mapping of the groundwater resources & development of ground water database for planning and development. During the financial year, the following achievements were realized under the project:

(i) 34 hydrogeological sites assessment was carried out to identify potential sites for drilling of exploratory boreholes, (ii) 4 Boreholes drilled in the following areas, Nchura, Narok and 2 boreholes in Kakibei and Kiboko, and, (iii) 3 aquifers cross-sections were profiled, namely; Nairobi aquifer suite, Mt. Elgon and Kakamega.

Figure 15: Map of borehole spatial distribution in Kenya

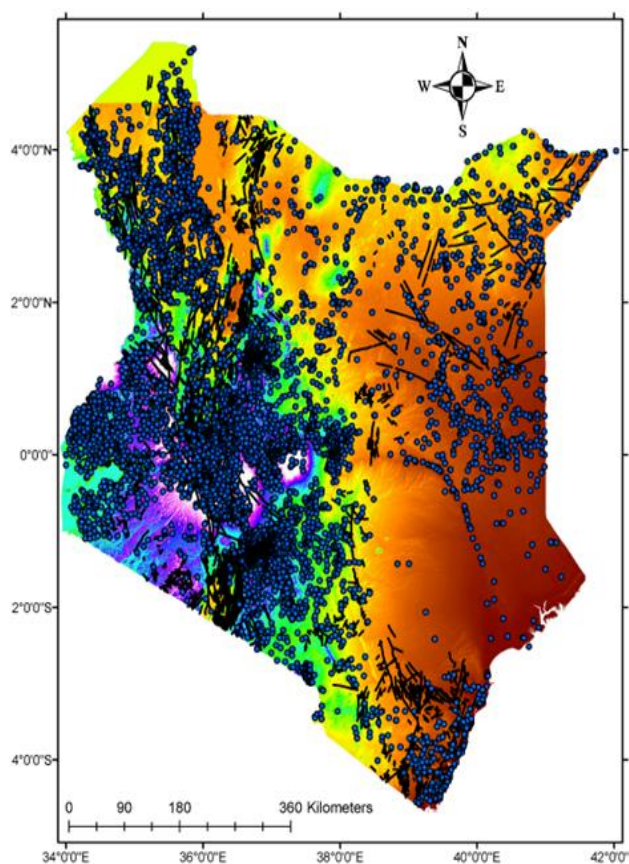


Table 3: Maintained Stations with Data

Basin Area	Station
LVNBA	<i>1AA01- Malaba, 1AH01 - Sio</i>
LVSBA	<i>1GD03-Nyando, 1HB04-L. Victoria, 1JA(New)-Ndoinet, 1JF04-Kipsonoi, 1JG04-Sondu, 1KB05-Gucha Migori, 1LA03-Nyangores,</i>
RVBA	<i>2B21-Turkwel, 2B33-Suam, 2EE07B-Perkerra, 2EG03-Molo, 2FC19-Njoro, 2GD06-L. Naivasha, 2K04-Ewaso Ng'iro,</i>
Athi	<i>3BA29-Nairobi, 3BC08-Ruiru, 3DA02-Athi at Munyu, 3F09-Athi at Kibwezi, 3KG01-Umba,</i>
Tana	<i>4BB01-Ragati, 4CA19-Karimenu, 4CC08-Thika, 4DA11-Thiba, 4DD02-Thiba, 4G01-Tana at Garissa</i>
ENNBA	<i>5BE01-Nanyuki & 5ED01-Ewaso Ng'iro at Archers post</i>

In addition, 1No. Surface Water monitoring station upgraded to telemetry in March 2023; i.e. RGS 3BA29, Nairobi River and configured to transmit data to WRA server

2.10 Water Quality Status

The Authority has an obligation to conserve and protect the water resources from pollution. The Authority continues to undertake water quality and pollution monitoring through her Water Quality Laboratory facilities on water samples from the rationalized monitoring network across all the six basins. The monitoring network encompasses point and nonpoint sources of pollution to the water resources. Due to emerging environmental challenges, new

appropriate monitoring stations can be established depending on the intentions that include research purposes.

Table 4: National Revised Water Quality Monitoring Network

Basin Area	Surface Water	Ground Water	Effluent Discharge	Total	1 ST Priority Stations
LVBA	44	19	39	102	24
LVBA	77	29	26	132	54
RVBA	49	30	23	102	27
Athi	55	45	32	132	44
Tana	58	19	27	104	30
ENNBA	46	19	21	86	23
Grand Total	329	161	168	658	202

Source: Water Quality Monitoring and Management Guideline. Volume D2 - 3, 2020

During the financial year under review, the following activities were carried out as part of conserving and protecting the water resources from pollution. Water quality assessment and assurance with the selected arid and semi-arid landscapes of Mid Tana, Sabarwawa and Chyulu Hills. This exercise was to generate data and information to appropriately advise on the water quality management interventions in view of limited availability in these landscapes. Water Abstraction and Pollution surveys in the Kiu river basin under Kiambu sub-basin and Suguroi river basin under Nanyuki sub -basin. Conducted of the findings and recommendations of the Water Abstraction and Pollution surveys that had been previously carried out in Mwanja Mbogo and Muringato river basins in

Nyeri and Moiben river basin in Eldoret for compliance with water resources regulations.

268 water quality and pollution monitoring stations were geo-referenced and mapped as part of the water quality and pollution monitoring network stations in the six basins. Carried out solid waste management assessment in the upstream catchment of Thwake dam. The key findings indicate solid waste as one of the major contributors to the decline of quality of water resources. Most of the solid wastes are on the riparian, road reserves and in the river course.

Solid waste management guidelines were developed and shared with the stakeholders - largely the County Governments, who are mandated to satisfactorily manage the solid waste. The solid wastes on the riparian reserves and in the river course. Quarterly water quality monitoring of the upstream ecosystems of Athi river feeding into Thwake dam within the Athi river Basin. Part of upstream of Thwake dam, up to Fourteen Fall. The spatial water quality during the period under review demonstrates the water quality trend along river profiles improved downstream due to river self cleansing - accelerated by turbulence in the river course, the aquatic flora and fauna that catalyzes biodegradation of organic pollutants. From the analyzed 2022 data, the results for BOD, COD, TDS and TSS are presented in figure 2-5. With respect to R.Ngong there is a general increase in concentration for all the parameters as the rivers flow from the outskirts of Ngong forest to Ngong at Kangundo road. This is quite evident as the river traverses the informal settlements of Kibera, Mukuru, Riverside and Soweto.

2.10.1 Results of Water Quality Monitoring

The average concentration of BOD, COD, TSS and TDS along the profile of R. Athi and its tributaries. The sampling locations for each river are arranged from the most upstream to downstream. Similar trend was observed in R. Nairobi. However, at Thika 14 falls, the BOD and COD concentration reduced significantly. The reduction can be attributed to (i) self-epuration of the river which is further facilitated by reduced pollutants input and aeration provided by the water falls albeit at low retention time, and (ii) dilution by tributaries

emanating from the Kiambu region which such as R. Ruiru which have less organic load. The two locations along R. Mathare, however, show no particular pattern even though the BOD and TSS are more or less constant. From Thika 14 falls to R. Athi at Thwake dam, the organic load reduces drastically by up to one order of magnitude, with the average BOD concentration dropping to between 4 and 7 mg L⁻¹. It should be noted that between 14 falls and R. Athi at Thwake dam, there is reduced input of pollutants from the few seasonal rivers that feed into the river. Moreover, the impact of urbanization on the water quality is also reduced. Like with the BOD, COD, TDS and TSS, the electrical conductivity showed a similar distribution pattern along each river profile.

Following the fishkill in Lake Victoria, regular water quality monitoring continued to be undertaken. The fishkill was associated to the lake upwelling phenomenon, a case where the low-oxygen water accompanied with debris on the lake bed comes up to the surface resulting into fish experiencing clogging of gills and suffocation leading to death

Conducted in collaboration with Thika River Basin stakeholders the Thika Water Allocation Plan. The plan entails sustainable water resources conservation, protection and use in an equitable manner amongst the stakeholders.

Over 584 water quality and pollution dataset were generated from the monitoring Programme of WRA in addition to the customers' samples that were received at the WRA laboratories for the analyses. Authority protects and conserves the limited freshwater resources from pollution through the Waste Disposal Control Plans that include Effluent Discharge Control Plans (EDCPs). In this case, all potential Effluent Dischargers are required to develop the prerequisite EDCPs, acquire an Effluent Discharge Permit (EDP) and adhere to therein conditions for compliance. The Water Resources Authority undertakes enforcement on EDPs to ensure that the effluent dischargers comply with the prescribed environmental and water resources standards.



3 WRA Performance

3.1 Water Allocation

3.1.1 Management of water use through the permitting system

The Authority allocates water through a permitting system, the permit classes are categorized as A, B C, D. A is for basic human needs and not motorized; this should be reported to WRA Sub Basin Area office for approval. This is important when calculating water balance. Class B is approved at the Basin Area and doesn't have much impact on the environment, Class C and D is advertised and thereafter C is approved at the Basin Area and D is approved at the Headquarter. Each category is within thresholds which vary per water body and basin.

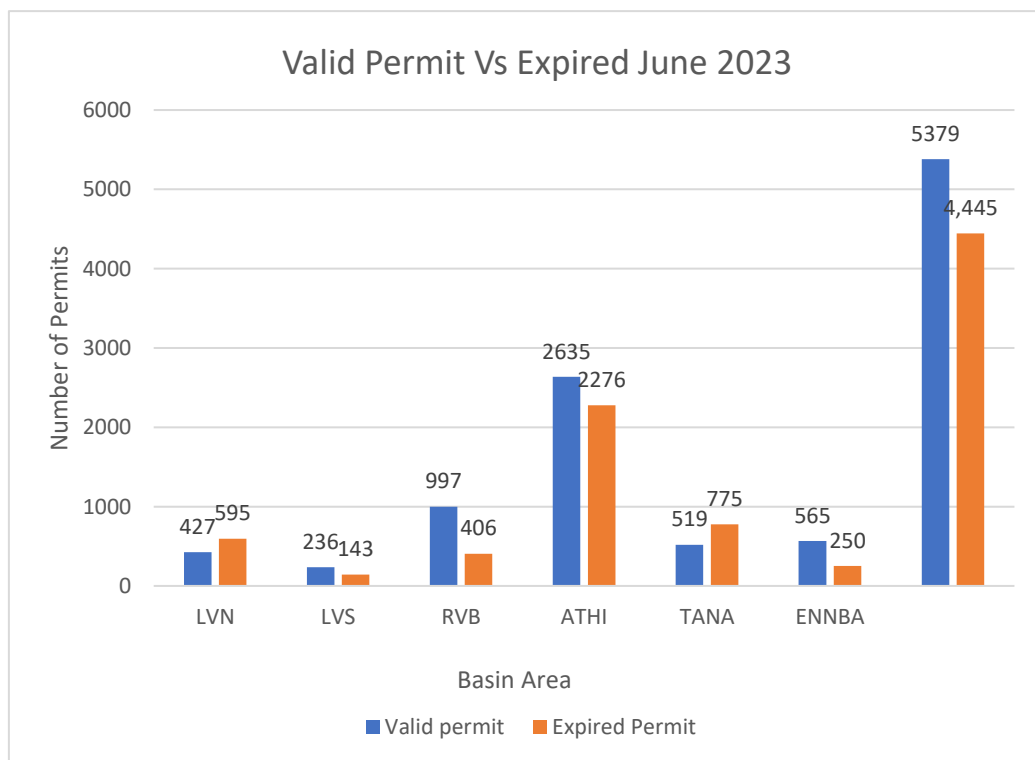


Figure 16: All Valid Vs Expired Permits

During the Financial year there were a total of 9,824 permits within the Authority in all the Basin Areas. 55 % were valid permits while 45% were expired permits. Athi Basin Area had the highest number of valid and expired

permits, the valid permits in Athi accounted for 49% of total valid permits of all the basins and the expired permits accounted for a total of 51% of all the expired permits in the six basin areas. This is related to the demand of water and Athi Basin has the two cities in it Nairobi and Mombasa with the highest populations. LVS had the lowest number of permits both valid and expired in comparison to all other basin areas.

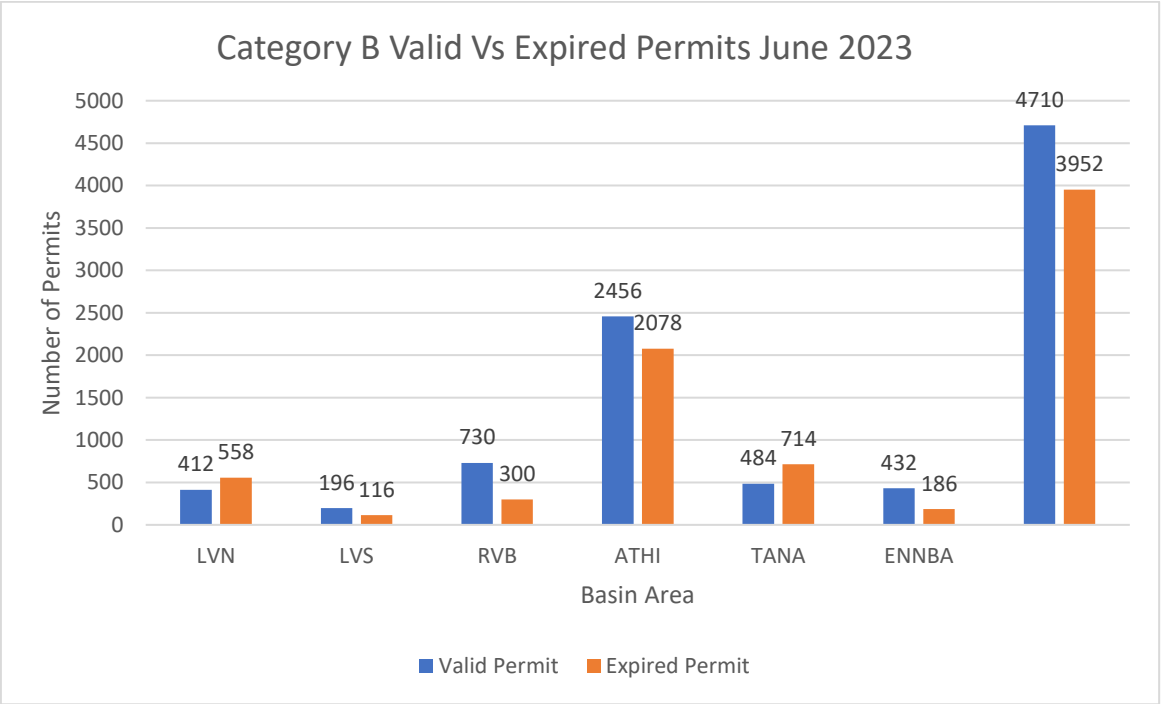


Figure 17: Category B Valid Vs Expired Permits

During the financial year there were a total of 8,662 category B permits in all the six basin areas, Athi Basin had the highest number of category B Valid permits at 52%, and the highest number of expired category B permits at 53%. The lowest percentage of valid permits was in LVSBA at 4% and the expired category B permits at 3%. Tana basin had the highest number of category B surface water abstraction permits owing to reliance on surface water. Generally, groundwater had the highest number of permits attributed to reliance on groundwater. The Authority intensified enforcement in all its basin areas and the sensitization of provisions of Water Resources Regulations 2021 and the access to e permit application process that the Authority introduced that facilitates easy access to the services.

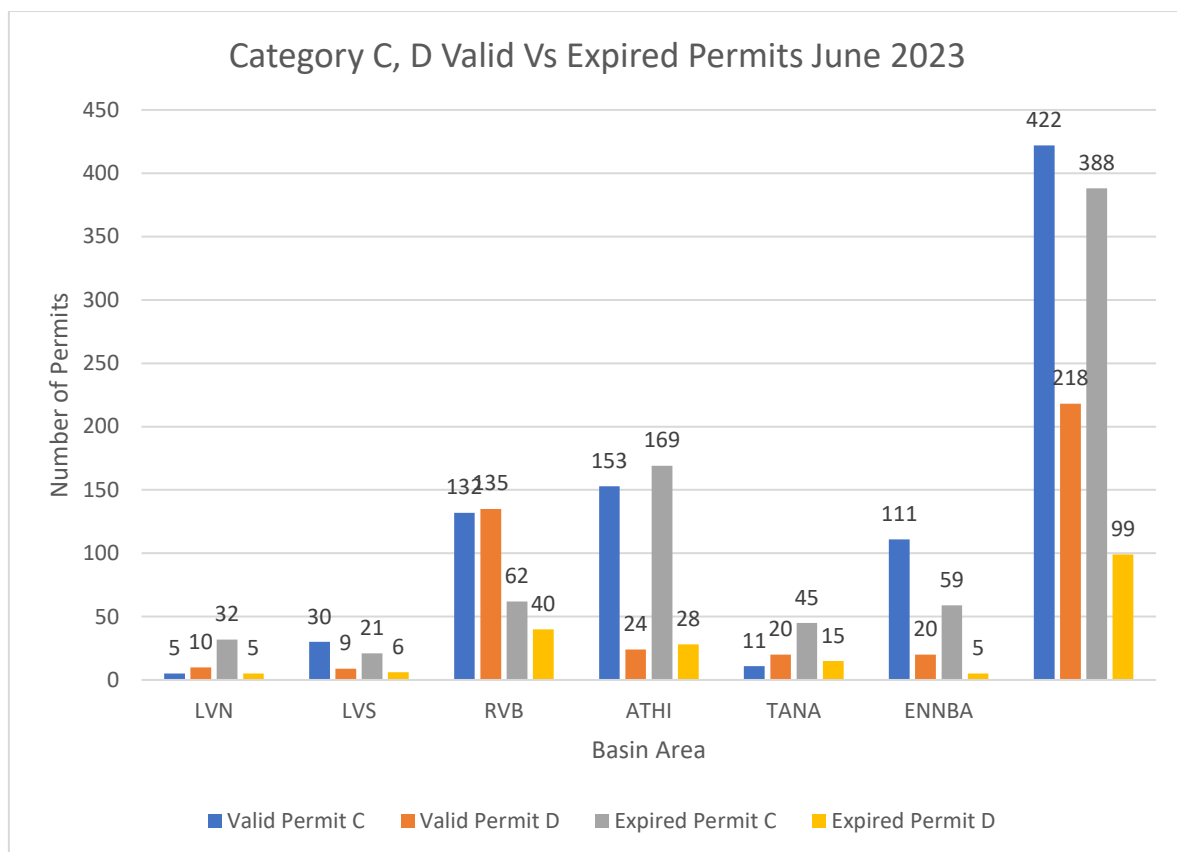


Figure 18: Category C, D Valid Vs Expired Permits

During the Financial Year there were a total of 800 Category C permits and 317 category D permits in all the six basin areas. 49 % of category C permits were valid and 69% category D permits valid. This was a significant improvement that demonstrates the ease and efficiency of the permit process through the online permit application. The Authority shall continue undertaking enforcement to permit processes and ensure the public accesses the online permit application system at all times. This facilitated the citizens abstracting water with permits and supports the Authority in data collection and equity in water allocation and informed decision making.

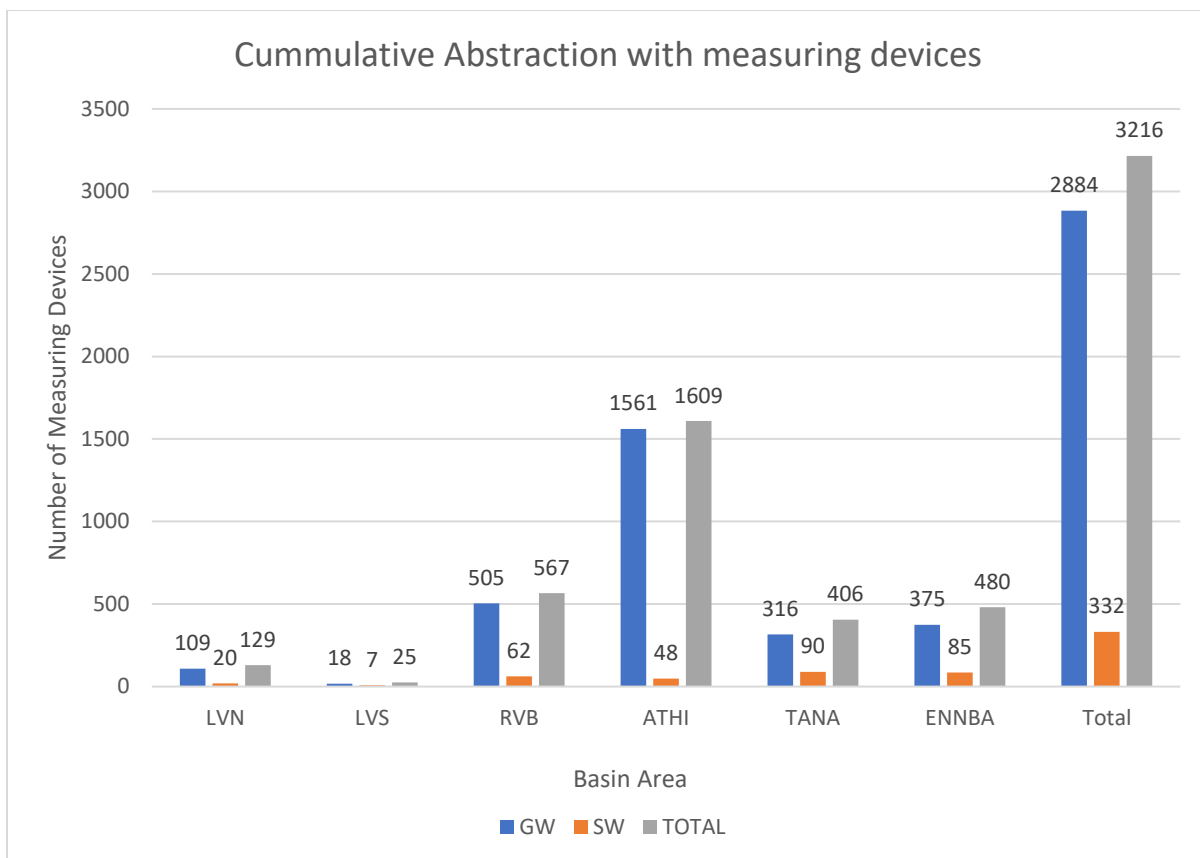


Figure 19: Cumulative Abstractions SW /GW with Measuring Devices

During the financial year there were a total of 3216 abstractions from surface water and groundwater with measuring devices. 90% of abstractions with measuring devices was groundwater and 10% was surface water. The Water Resources Regulations provide for installation of smart meters by all abstractors and thus there were efforts by the Authority to sensitize its customers on the move from master meters to smart meter installation. The smart meters ensure accounting for water allocated and use and the correct billing that promotes sustainability and efficiency in water use. The Authority shall continue with its awareness creation and enforcement on the master meters and thus will ensure that water is always available and there is minimum and reduced violation of reserve flow.

3.2 Water Allocation Criteria

3.2.1 Water use by category

Table 5: Water Use by Category

Volume of water by category of water use up to June 2023 (x 1000 m ³ /day)														
Basin	Public		Domestic		Livestock		Irrigation		Industrial		Power		Other	
	SW	GW	SW	GW	SW	GW	SW	GW	SW	GW	SW	GW	SW	GW
LVN	134.62	0.9325	5526.04	16.78	2.03	0.235	11.4	0.30	224.45	0.232	411.0	0	12.466	0.18
LVS	104.84	0.232	7.192	6.521	0.102	0.085	17.546	0.306	24.635	0.32	28366.644	0	1.906	0.09
RV	13.804	65.111	18.518	26.464	3.741	2.479	100.152	268.269	12.174	3.334	1658.509	1.52	13.06	18.364
Athi	88.788	60.364	12.141	111.543	0.165	2.819	299.482	33.984	38.71	33.91	0.07	1.48	1.06	1.854
Tana	408.68	107	490.943	108.184	3.829	1.82	952.148	2.2	12.2	0.16	806.36	0	60.15	1.1
ENN	31.233	2.241	61.898	10.52	8.323	1.38	116.92	228.693	1.068	0.18	0.07	4.4	5.715	00

Table 6: Status of Surface Water Storage

	<i>PANS</i>	Class of Dam as at June 2023		
		<i>CLASS A</i>	<i>CLASS B</i>	<i>CLASS C</i>
No. of applications	0	68	10	5
Valid authorizations	11	25	6	0
Expired Authorizations	5	149	129	18
Valid permits	1	20	28	5
Expired permits	3	13	349	6
Valid+ expired permits	4	33	377	11

3.3 Engagement of WRUAs in Catchment Conservation and Restoration

3.3.1 Status of WRUA

WRUAs are established for purposes of collaborative management of water resources and conflict resolution. The WRUAs therefore provide the platform for water users and other stakeholders to collaboratively manage the water resources. There are currently 760 registered WRUAs in Kenya as at June 2023 against the optimal number of 1237 WRUAs. WRUAs prepare Sub-catchment management plans to identify water resources issues in their sub-catchments and identify solutions and interventions for addressing the water resources challenges such as conflicts, water quality and catchment degradation, climate change, water scarcity among others. The Authority partners with various development partners to support WRUAs to implement SCMP activities. Additionally, 16no. New sub catchment management plans were developed and 22 no. WRUAs were reviewed in the FY 2022/23.

3.3.2 Kenya National Association of Water Resources Users Associations (KeNAWRUA)

The Kenya National Association of Water Resources Users Associations (KeNAWRUA) is a member organization bringing together all individual registered Water Resources Users Associations (WRUAs) in Kenya. The idea of establishing KeNAWRUA emanated from the WRUAs who felt the need to have a national body which would represent their interest, be the voice lobby and mobilize resources for cooperative management of water resources and conflict management. KeNAWRUA was registered with the Registrar of Societies in October 2021. The main purpose of KeNAWRUA is to represent the interest of WRUAs nationally, regionally and globally, to advocate and lobby for the sustainable management of water resources, mobilize resources for collaborative management of water resources and conflict management and promote good governance within the WRUAs. The WRUA has an established office in Nairobi with an interim executive officer who oversees the day-to-day activities of the Association, in which in the next financial year advertisement for the position shall be undertaken, the Association currently has 42 registered

members. Since its establishment, KeNAWRUA has partnered with WWF to train its members in IWRM at the Egerton University including several youths with whom they would like to encourage to participate in water resources management and conservation activities, undertaken training for its members on understanding the Water Resources Regulations (WRR) 2021 and practical lessons training for its members community science and simple water quality testing.

Table 7: Status of WRUAs and SCMPs

Basin	WRUA Formation Status			SCMP Developed			SCMP Implementation	
	Potential	Achieved	Proportion%	Potential	Achieved	Proportion %	Total	Proportion %
LVNBA	106	94	89	106	73	69	5	7
LVSBA	137	107	78	137	51	37	11	22
RVBA	175	88	50	175	52	30	6	12
Athi	240	188	78	240	91	38	8	7
Tana	309	176	57	309	94	30	11	12
ENNBA	270	107	40	270	72	27	5	7

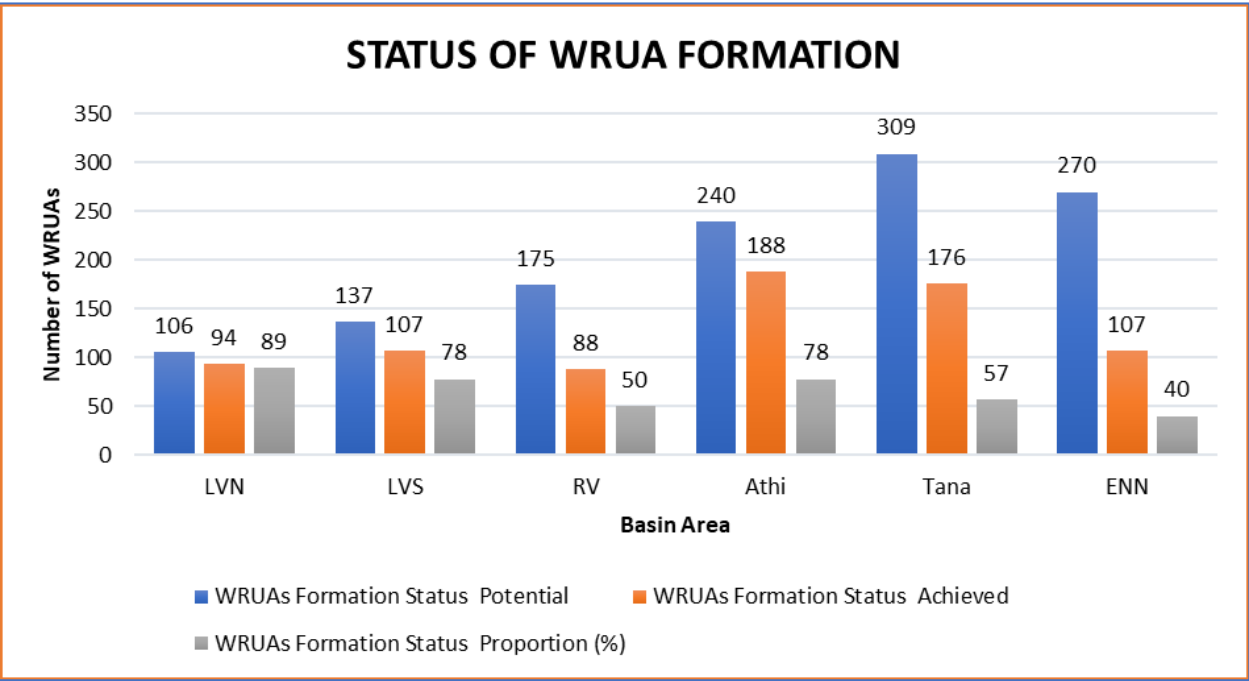


Figure 20: Status of WRUA Formation

The optimum number of WRUAs to be formed in the country by WRA is 1237. As at June 2023 there were 760 WRUAs established cumulatively. Establishment and operationalization of WRUAs plays a key role in conflict management by providing a platform for communities and stakeholders to collaboratively manage water resources and manage water related conflicts.

The data displayed above shows the number of potential WRUAs per basin area and the achievement towards realizing this potential. During the year under review the status of WRUA formation was as follows; LVN Basin achieved 89%, LVS Basin achieved 78%, Rift Valley Basin had achieved 50%, Athi Basin had achieved 78%, Tana Basin had achieved 57% and Ewaso Ng’iro North Basin had achieved 40%. Lake Victoria North therefore has the highest number of WRUAs formed (89%) compared to Ewaso Ng’iro North which had the lowest WRUA formation (40%).

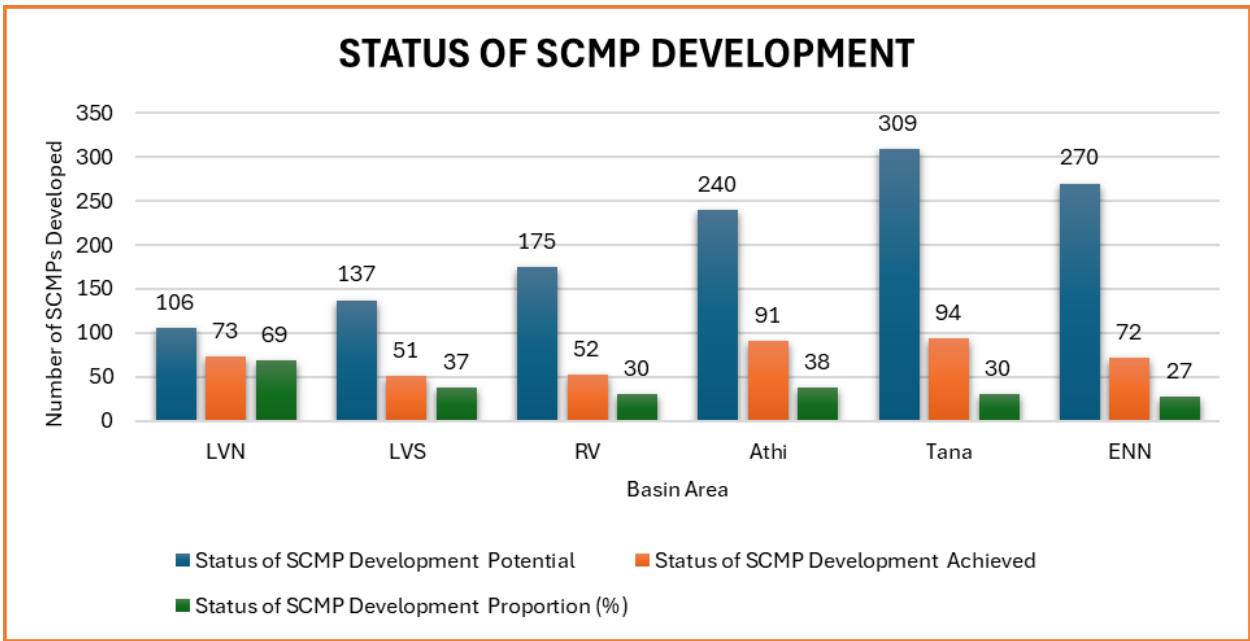


Figure 21: Status of SCMP Development

WRA provides technical support to the WRUAs to prepare Sub-Catchment Management Plans (SCMPs). During preparation of SCMPs, science and local knowledge is harmonized to map out and analyze water resources issues in the sub-catchments, to arrive at interventions which are implemented to solve the identified issues. The SCMPs are used as a shopping basket for WRUAs to seek

for funds to implement the unique water resources issues for each sub-catchment.

The data above shows the numbers and % age of SCMPs cumulatively developed to date in comparison to the potential number to be achieved. During the period under review, LVN Basin had achieved 69% in development of SCMPs, LVS basin had achieved 37%, Rift valley basin had achieved, 30%, Athi basin had achieved 38%, Tana Basin had achieved 30% and Ewaso Ng'iro North Basin had achieved 27% in development of SCMPs out of the total potential number to be achieved for the basin area. This meant that LVN Basin had the highest %age of SCMPs developed (64%) compared to Rift Valley and Tana Basin areas which were at 30% each.

WRUAs Develop SCMPs with the financial support of the Authority and/or other stakeholders. The Authority has an elaborate framework and manual to guide the development of SCMPs.

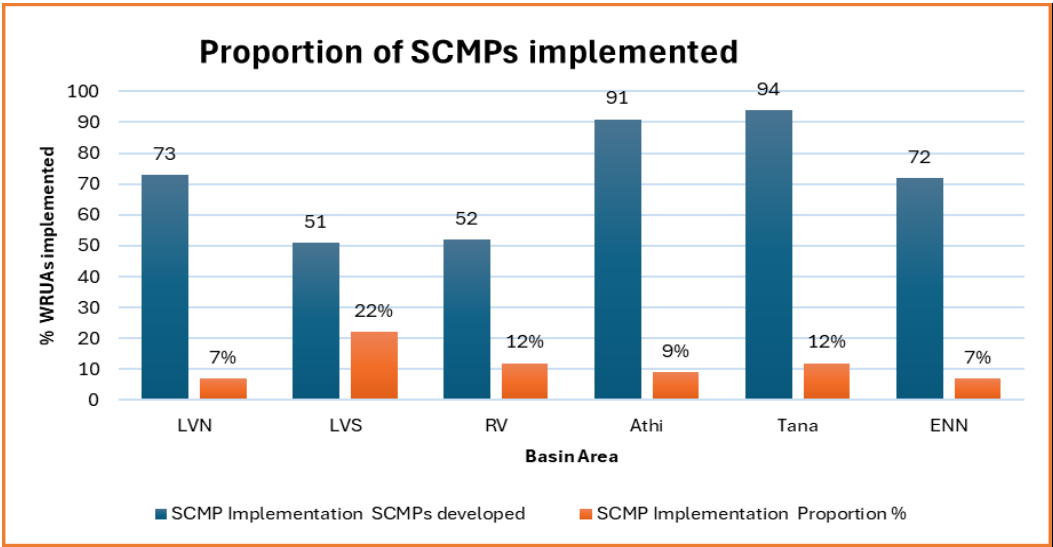


Figure 22: Proportion of SCMP Implemented

Implementation of SCMPs is important in ensuring the WRUAs address some of the water resources challenges facing their sub-catchments. This helps to improve catchment conditions, increasing water availability, protection of vulnerable water ecosystems, improvement in water quality and quality and improved adaptation to the effects of climate change among other benefits.

The data above shows that during the period under review, LVN Basin 7% of SCMPs were implemented, in LVS 22% of SCMPs were implemented, in RV Basin 12% SCMPs were implemented, in Athi Basin 9% SCMPs were implemented, in Tana Basin 12% SCMPs were implemented and in ENN Basin 7% SCMPs were implemented. The percentage proportion was based against the SCMPs developed vis a vis the SCMPs implemented per basin. WRUAs implement SCMPs with the support of various partners and stakeholders. Implementation of SCMPs vary each year depending on availability of funds.

3.4 Water Resources Assessment and Monitoring

3.4.1 Groundwater Potential and Exploitation

During the period 2022/2023, the country continued to experience drought, the hardest hit counties were in the Northern and North Eastern areas of the country. Horn of Africa groundwater for Resilience Project commenced the process of assessment of eight (8 No.) aquifers to inform groundwater investments in exploitation, conservation and managed aquifer recharge of groundwater resources to ensure sustainability in Turkana, Marsabit, Wajir, Mandera and Garissa Counties which will continue to the financial year 2028/2029. Through the same initiative process of developing groundwater information systems has commenced for information acquisition, storage and sharing of groundwater data for decision making. In general, annual groundwater potential of the country was estimated and the map presented in figure 13. Towards sustainable utilization of groundwater resources water allocation plan for Nairobi Aquifer Suite has been reviewed and also the water allocation plan for Lamu Island has been developed. Concerted efforts have also been employed in determination of the abstraction boreholes through development and population of the WRA borehole database across the country. A new groundwater information system is being proposed to be rolled out under the Horn of Africa Groundwater for Resilience Project

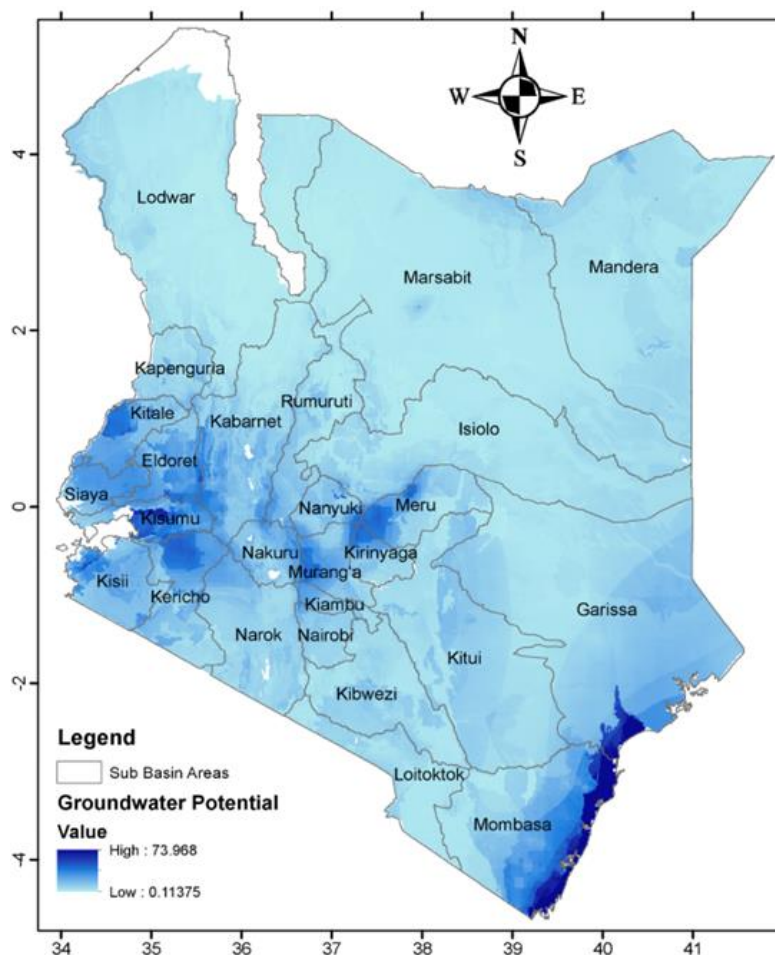


Figure 23: Groundwater Potential Map (2019)

Other initiatives are underway to establish groundwater potential at Basin and aquifer level; Ministry of Water Sanitation and Irrigation and United States Geological Survey (USGS) Mapping groundwater potential in Mandera County and Athi and Tana basin respectively. Tana and Athi basin assessments constitute 17 counties. UNICEF is mapping groundwater resources in Turkana, Marsabit, Garissa, Mandera and Marsabit.

3.4.2 Groundwater Monitoring

The Water Resources Authority has been undertaking groundwater resources monitoring since its establishment in 2005. This has mainly been an ad hoc arrangement with owners of production wells for manual observations. WRA

has continued to get data from the private boreholes and its own dedicated monitoring stations across the country. Currently WRA has established 187 groundwater monitoring stations across the country. However, it is crucial that dedicated groundwater monitoring networks are set up across aquifers to enable monitoring of groundwater levels and quality. In 2022/2023 the Authority has continued to maintain the existing telemetric network which are providing observation data to inform groundwater levels under the Nairobi aquifer suite, Tiwi aquifer and the Lamu sand dunes aquifer. Seven monitoring wells in the Lamu sand dunes and two in Nairobi aquifer suite were completed in the year under review. 5 data loggers are in place in Kisumu, Kabatini and Bungoma monitoring boreholes. These monitoring wells are already giving groundwater level data received in the water resources information system of WRA. Through the Horn of Africa – Groundwater for resilience project, it is proposed to install further monitoring wells in the seven aquifers; Walda Rawana, Lotikipi, Napuu, Daua, Merti, Logologo Shuur and Neogene aquifers. This will lead to better groundwater management towards goals of ensuring sustainable access and management of groundwater.

Table 8: Groundwater monitoring stations per basin per aquifer class

	Strategic	Major	Minor	Special	Total	% Operational
Arhi	44	12	4	1	61	53.3
ENNCA		16			16	75
LVN		20			20	90
LVS		14			14	64.3
RVCA	27	20			47	85.1
Tana		22		7	29	85.7
Total	71	103	4		187	75.6

3.4.3 Surface Water Monitoring

The attainment of the government's target of ensuring improved availability and access to water by all requires a robust and comprehensive assessment and monitoring system which accurately tracks quantity and quality of water resources for effective management. Monitoring networks in the country comprises meteorological stations and hydromet stations.

There are a total of 223 rationalized surface water monitoring stations categorized as National, Management Unit, Intra Management unit and Special Stations. The stations have been geo-referenced and uploaded in the National Monitoring and Georeferenced Information System. The number of stations, operational status, distribution of hydro meteorological and weather monitoring stations for the country are shown in table below

Table 12: Surface Water Monitoring Stations and Operational Status

Basin Area	Category				Total	% Operational
	National	Management Unit	Intra Management Unit	Special		
LVBA	5	8	9	6	28	82.6
LVBA	5	14	20	1	40	75
RVBA	7	13	20	0	40	82.5
Athi	3	4	22	3	32	75
Tana	1	7	21	16	45	80
ENNBA	1	4	31	2	38	81.5
Total	22	48	122	31	223	

3.4.5 Rehabilitation of the Surface Water Monitoring Station

Surface Water stations are periodically rehabilitated and upgraded to telemetry to improve data collection and decision making. The upgrading of stations to automatic logging and telemetric transmission has ensured that accurate and timely data is captured. The automatic stations are configured to record data on hourly basis and transmit to a server which can be accessed via web-based portal.

Water Resources Authority rehabilitated 4No. surface water monitoring stations and the Ministry 4No. to telemetric stations under the IGAD HYCOS project, totaling to 8 telemetry stations. The upgraded stations by WRA to telemetric stations were the ones whose sensors had been washed away by floods. The work done included reinstallation of water level sensors,

configuration, calibration and reactivation of the loggers and modems to transmit data to the WRA server. The 8 rehabilitated stations were as follows:

- I. 1AA01 Malaba River in Lake Victoria North Basin area
- II. 2B33 Suam River and 2B21 Turkwel River in Rift Valley basin area
- III. 4BB01 Ragati River in Tana basin area
- IV. 5ED01 EwasoNg'iro in EwasoNg'iro North basin area
- V. 2EC02 Rongai in Rift Valley Catchment
- VI. 5DA08 Isiolo in Ewaso Ng'iro North Basin area
- VII. Kirichwa Kubwa in Athi Catchment

Re installation of the water level sensor at RGS 2B21 Turkwel river	Malaba RGS 1AA01 after installation	Reconfiguration, Calibration and reactivation of the sensor at RGS 2B33 Suam river
---	-------------------------------------	--

3.4.6 Weather Monitoring Stations

Water Resources Authority operates a weather monitoring network comprising rainfall, evaporation and climate stations. A total of 156 rainfall stations, 47 evaporation and 17 climate stations were operational.

Table 9: Weather Monitoring Network

Basin Area	Rainfall	Operational	Evaporation	Operational	Climate	Operational
Athi	45	25	7	3	5	2
ENN	26	6	4	2	9	2
LVN	72	30	10	3	5	0
LVS	47	36	20	18	12	10
RV	23	21	7	6	3	3
Tana	53	38	17	15	2	0
Total	266	156	65	47	36	17

The weather monitoring stations operated by WRA are strategically located at various parts in the catchments. Data is also obtained from various stakeholders such as private institutions, government and learning institutions. The data collected from these stations is used for modeling and planning purposes. 45 weather stations were configured to transmit data to the WRA server. The stations were upgraded to telemetry through the Kenya Water Security and Climate Resilience Project and had been configured to only transmit data to Kenya Meteorological Department server. The proposed configuration enabled data to be transmitted to WRA sever as backup.

Table 10: Distribution of Monitoring Stations

Basin Area	Surface Water Station	Station with water level records	Stations with updated rating curves (earmarked for FY 2022/2023)	Monitoring boreholes	operational boreholes	% Operational	Boreholes with Water Rest data
LVN	28	22	2	20	18	90	18
LVS	47	28	2	37	7	23	33
RV	42	17	2	55	32	41.82	32
Tana	45	30	2	45	23	51.1	18
Athi	47	40	2	25	20	80	20
ENN	40	35	2	16	12	75	12

3.4.7 Maintenance of Reserve Flows

3.4.7.1 Rating Curve Update

12 Rating curves were updated for the following stations the following regular gauging stations;

- *1GD03 Nyando, 1AH01 Sio River, 1KB05 Kuja Migori River, 1FG02 Yala River, 2EE07B Perkerra River, 2B33 Suam River, 3DA02 Athi River at Munyu, 3GA02 Tsavo River, 4BE01 Maragua, 4G01 Tana River at Garissa, 5BE01 Nanyuki and 5ED01 Ewaso Ng'iro River.*

3.4.8 Water quality monitoring network

The network aims to generate water quality and pollution data with respect to ambient, point (specific site) source and nonpoint (diverse/diffuse) sources of pollution to the water resources. Some of the water quality and pollution monitoring stations totaling 658 coincides (Table 1) with the established 223 surface river gauging stations (RGS) and 154 groundwater monitoring boreholes. This is intended to ensure that water quantity and quality are monitored simultaneously in order to establish the availability of water of good quality and quantity. Most of the stations are manual but a few are prioritized for automation or telemetric systems to relay real-time data to the database.

Table 11: National Water Quality Monitoring Network

Basin Area	Surface Water	Ground Water	Effluent	Monitoring Station	Priority Stations (for automation)
LVNBA	44	19	39	102	24
LVBSA	77	29	26	122	54
RVBA	49	30	23	102	27
ABA	55	45	32	132	44
TBA	58	19	27	104	30
ENNBA	46	19	21	86	23
Total	329	161	168	658	202

3.4.9 Water quality trend on case study rivers

Generally, the surface water bodies traversing urban and peri-urban areas are more susceptible to pollution when compared with ground water bodies. Further, the surface water bodies are of better quality in the upstream of settled town areas. The water quality deteriorates as the surface water bodies navigates through settled informal and formal areas, open markets, Jua kali zones and industrial areas. However, to some extent, the river waters improve due to self-cleansing with contributions accelerated by river morphology, turbulences, aquatic flora and fauna biological activities.

Case study 1: Water quality trends in Athi rivers upstream of Thwake dam

A. pH and conductivity

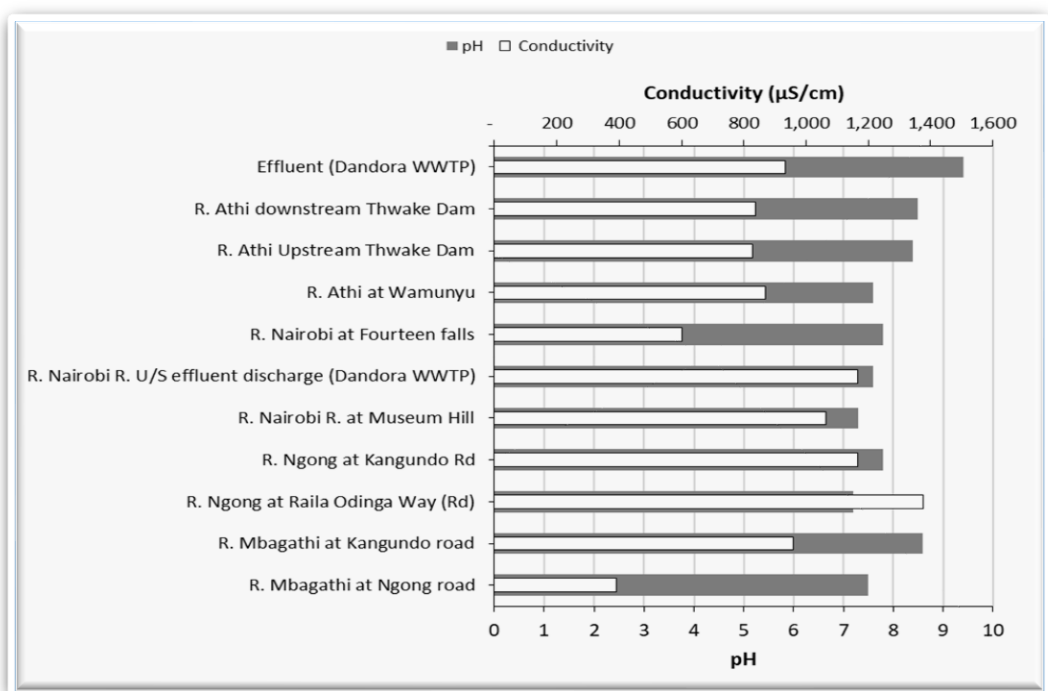


Figure 24: pH and Conductivity trend in R. Athi and its tributaries

B. Solids

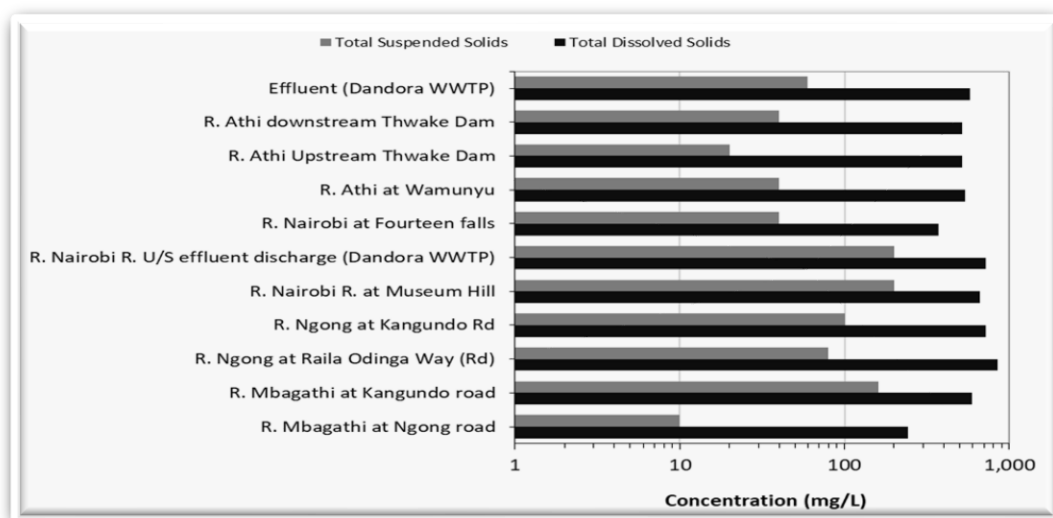


Figure 25: TDS and TSS trend in R. Athi and its tributaries

C. Organic matter

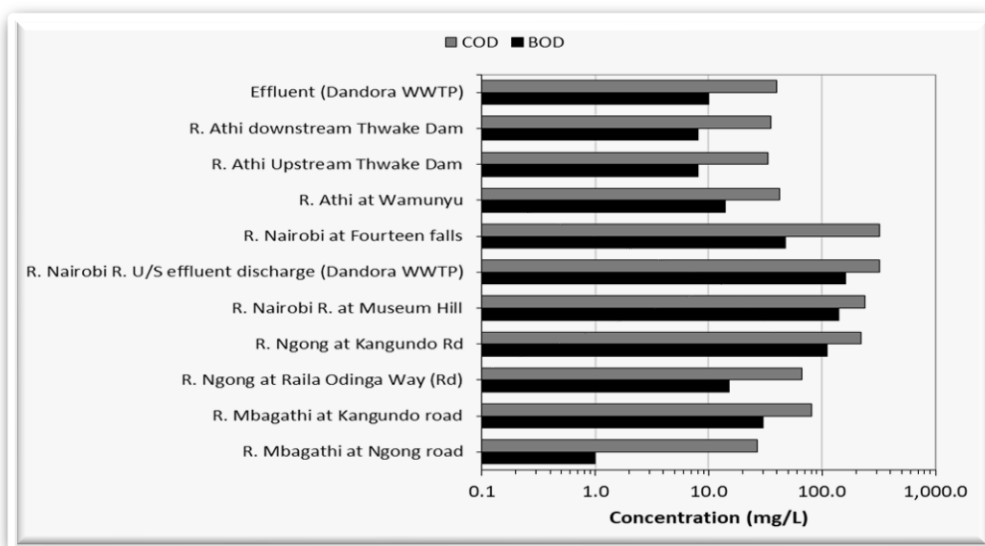


Figure 26: BOD and COD trend in R. Athi and its tributaries

D. Nutrients

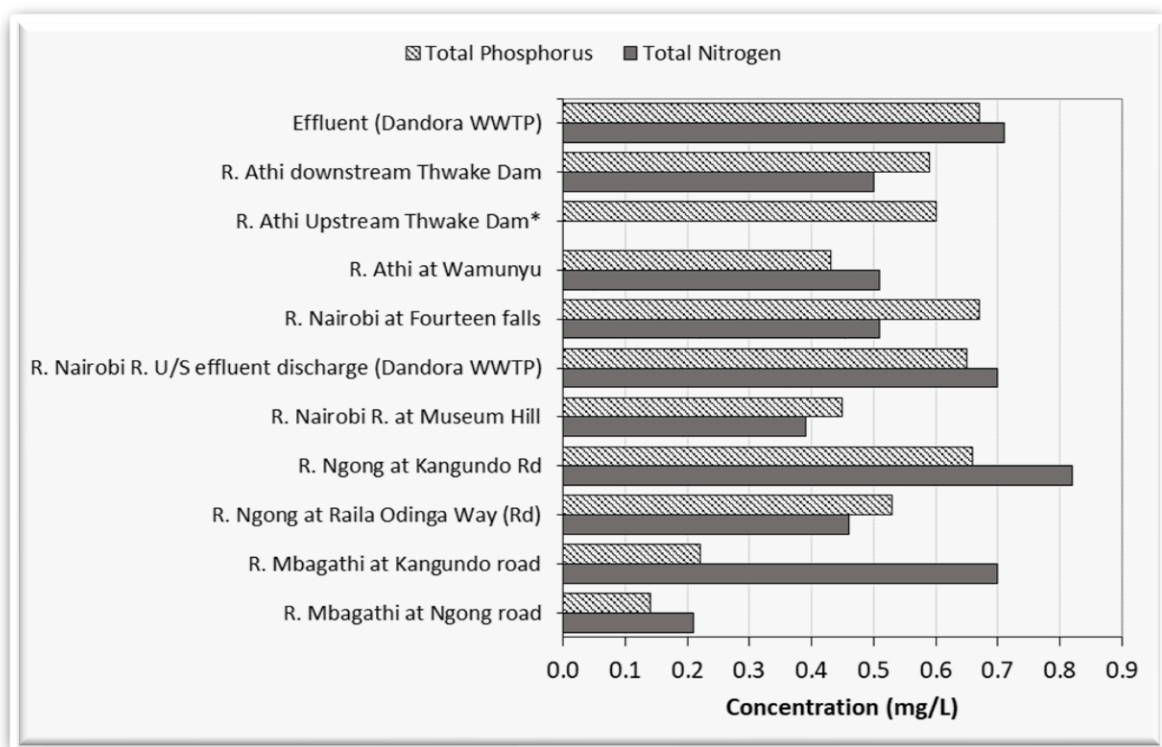


Figure 27: Concentration of Total Nitrogen and Total Phosphorus in R. Athi and its tributaries * concentration of Total Nitrogen (3.04 mg L⁻¹) is an outlier

E. Heavy metals

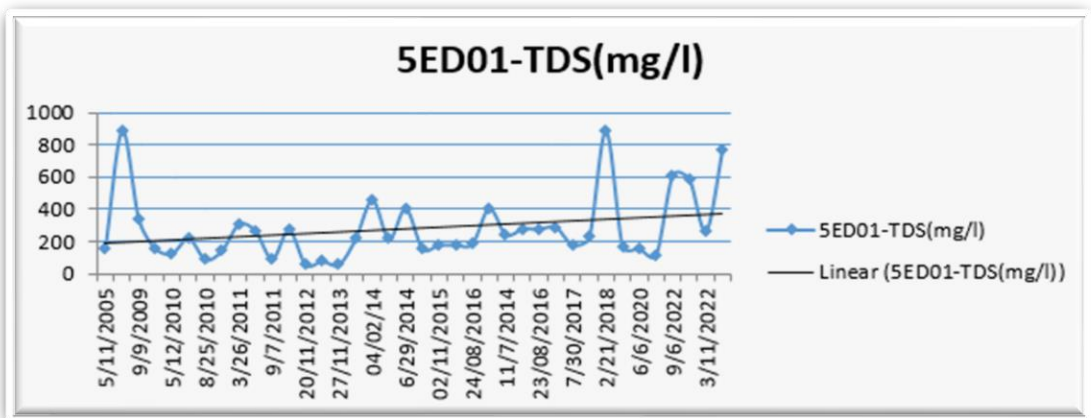
Table 12: Concentration (mg L⁻¹) of selected heavy metals in R. Athi and its tributaries. ND: Not detected

Sampling site	Lead	Cadmium	Zinc	Chromium
R. Mbagathi at Ngong road	ND	ND	ND	ND
R. Mbagathi at Kangundo road	ND	ND	0.10	ND
R. Ngong at Raila Odinga Way (Rd)	ND	ND	0.02	ND
R. Ngong at Kangundo Rd	0.05	ND	0.40	ND
R. Nairobi R. at Museum Hill	ND	ND	0.20	ND
R. Nairobi R. U/S effluent discharge (Dandora WWTP)	ND	ND	0.20	ND
R. Nairobi at Fourteen falls	0.06	ND	0.50	ND
R. Athi at Wamunyu	ND	ND	1.60	ND
R. Athi Upstream Thwake Dam	ND	ND	0.01	ND
R. Athi downstream Thwake Dam	ND	ND	1.50	ND
Effluent (Dandora WWTP)	ND	ND	0.30	ND
Mean	0.06		0.50	
Min	0.05		0.01	
Max	0.06		1.60	

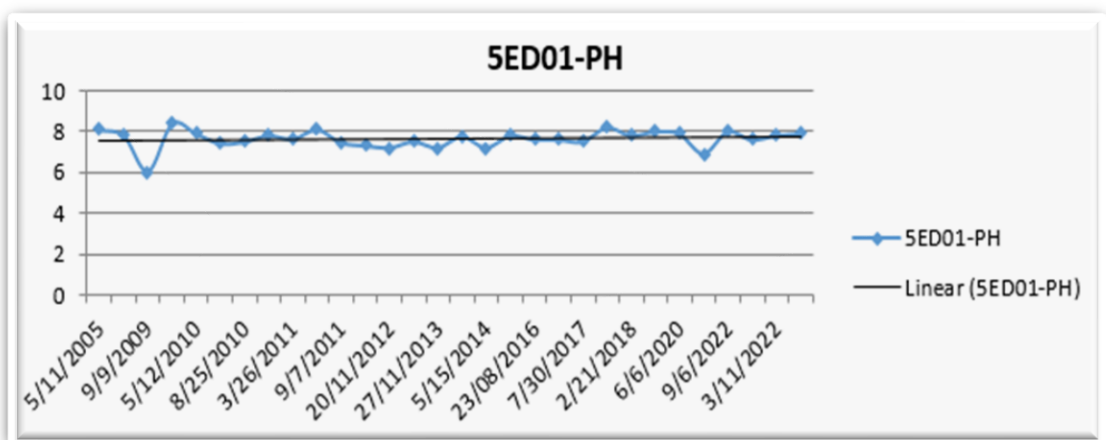
Case study 2: Water quality on Ewaso Ng'iro River

Table 13: Water Quality on Ewaso Ng'iro River

Station	Data sets	Turbidity for the last monitoring period (NTU)	Long Term Mean (NTU)	Maximum Recorded Value (NTU)	Minimum Recorded Value (NTU)
5ED01(Achers Post)-National	28	3.32	716	6180	3.32
5DA07 (Isiolo)-MU	22	42	66	786	0
5BC04(Ewaso ng'iro)-MU	41	628	164	798	13.51
5BE20(Nanyuki)-MU	34	20	64	545	0
5DC01(Ewasong'iro)-MU	4	41.35	64.6	164	4.78
5AC08 (Ewasonarok)-MU	5	20	64	120	0.22



The above graphs represent Ewaso Ng'iro River at 5ED01 (national station) with respect to conductivity and TDS, there is an empirical correlation between the two parameters.



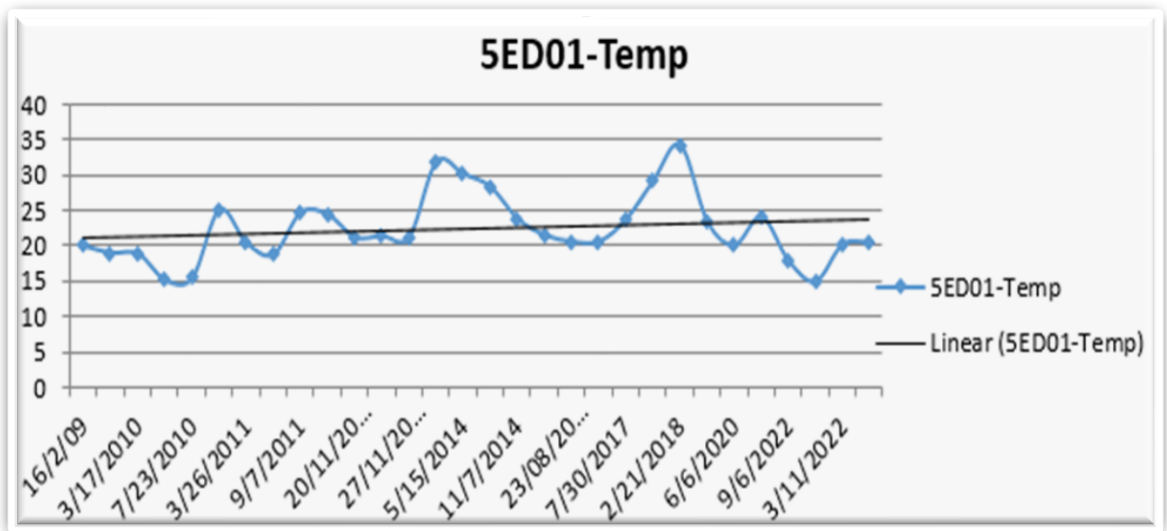
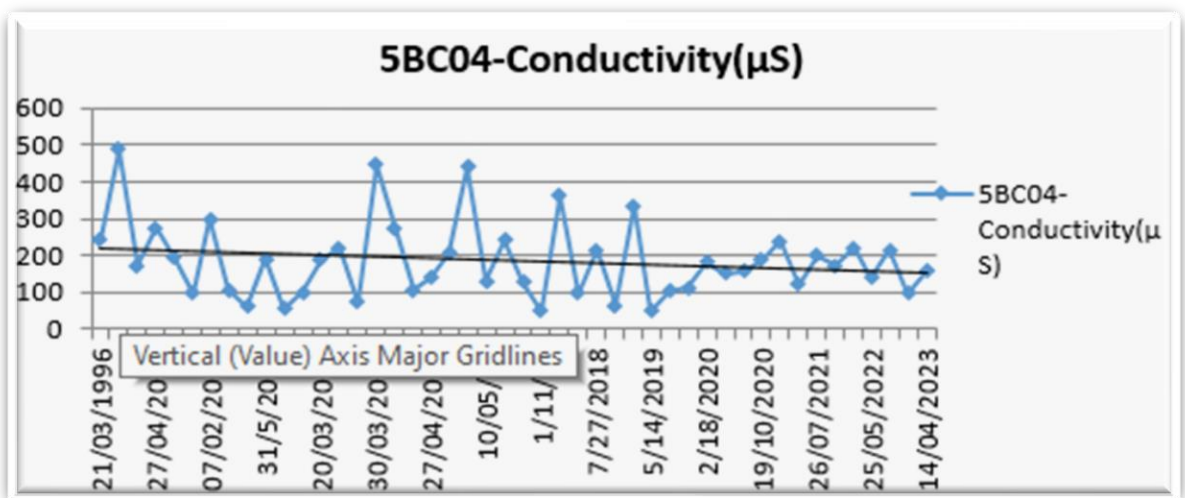
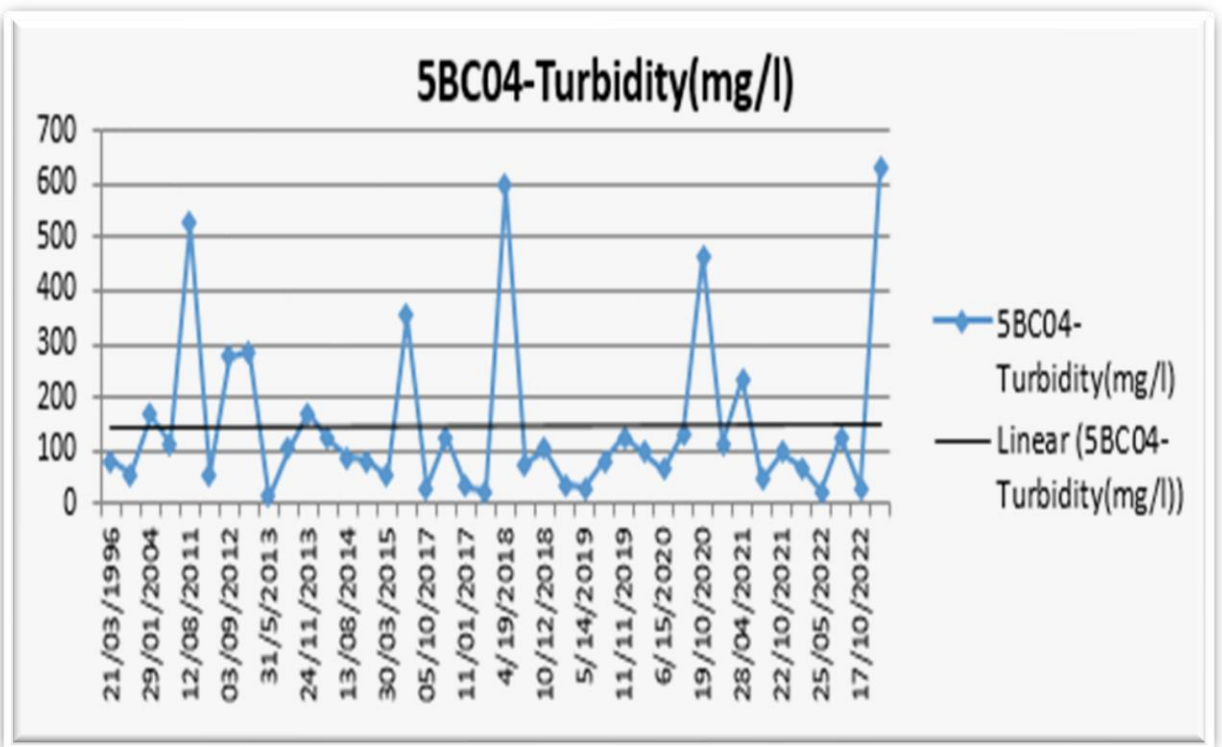
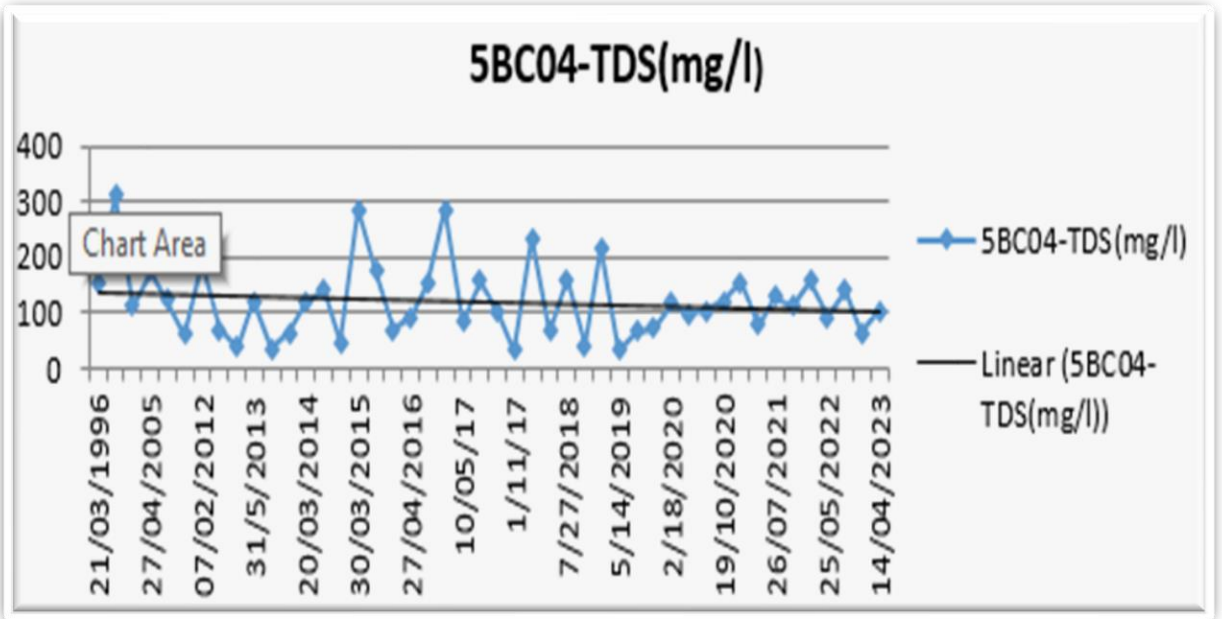


Figure 28: Ewaso Ng'iro River at 5BC04





Turbidity exhibits a slightly increasing trend, due to high peaks in 2011, 2018 and 2023 (<500NTU). Increased rains result in decreased conductivity and an

increase in turbidity and the contrary is also true with other factors such as catchment conditions being kept constant. This station monitors part of Aberdare and part of Mt. Kenya River systems. The trend in both conductivity and TDS are insignificantly declining with time without sharp peaks between monitoring periods for the last five years.

Table 14: Broad water pollution threats in the basin

Basin	Municipal wastes/ Untreated sewage	Industrial Waste Discharges	Soil erosion/ Sediment	Mining wastes e.g Gold, Titanium, building stones and sand	Oil drilling wastes, petroleum product spillages	Agro-chemicals
LVN	✓	✓	✓	✓	✓	✓
LVS	✓	✓	✓	✓	✓	✓
RV	✓	✓	✓	✓	✓	✓
Athi	✓	✓	✓	✓	✓	✓
Tana	✓	✓	✓	✓	X	✓
ENN	✓	X	✓	✓	X	✓

3.5 Economic and Financial management

3.5.1 Improving water use efficiency and economic value for water

Table 15: Status of water use charges June 2023

Basin	Amount of revenue (Million KES) by June 2023				Efficiency (%) by June 2023	
	Potential Kshs. (M)	PC Target Kshs. (M)	Billed Kshs. (M)	Actual collection Kshs. (M)	Water use (Percentage)	Revenue collection (Percentage)
HQ	355.2	206.67	260	189.8	73%	92%
LVN	53	30.84	100.004	39.98	189%	130%
LVS	51	29.67	83.902	43.18	165%	146%
RV	144	83.79	185.93	159.3	129%	190%
Athi	205	119.28	399.261	273.5	195%	229%
Tana	158	91.93	94.712	99.9	60%	109%
ENN	65	37.82	53.17	65.24	82%	173%
Total	1,031.20	600	1176.979	870.9	114%	145%

‡Water use efficiency = Bills against potential: Revenue collection efficiency = Actual collection against target

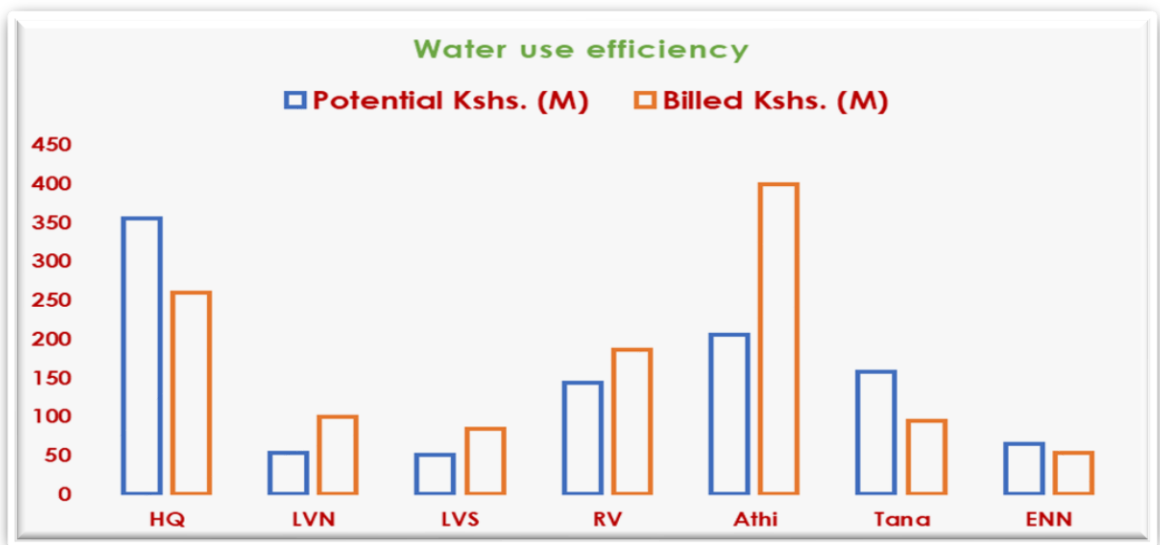


Figure 29: Water Use Efficiency

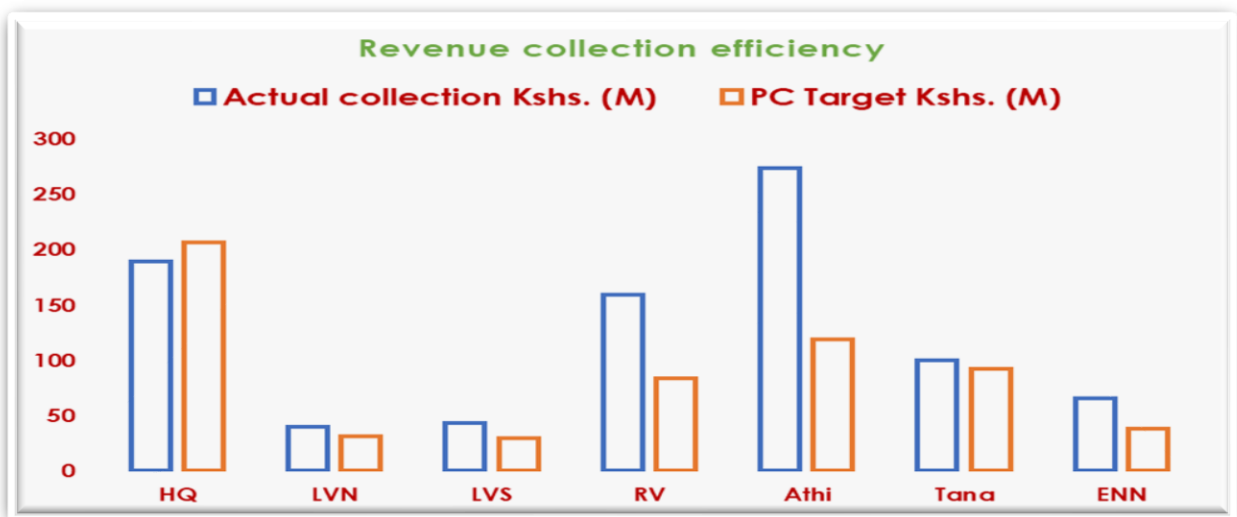


Figure 30: Revenue Collection Efficiency

3.6 Information Management

3.6.1 Managing water resources monitoring data and information

Table 16: Water Resources Monitoring Stations and data frequency June 23

Basin Area	Operational stations			Duration of data collection		Frequency of update
	SW	GW	WQ	SW	GW	
LVN	28	20	102	12 months	12 months	Monthly
LVS	40	14	132	12 months	12 months	Monthly
RV	40	47	102	12 months	12 months	Monthly
Tana	45	29	104	12 months	12 months	Monthly
ENN	38	16	86	12 months	12 months	Monthly
Athi	32	61	132	13 months	12 months	Monthly

SW: surface water; GW: groundwater; WQ: water quality

3.7 Gender in WRA

3.7.1 Gender Representation in WRA

The total number of staff in WRA are 629 out of which 57% are Males and 43% are females. The Authority has gender policy in place and is deliberate on undertaking gender mainstreaming in its work policies.

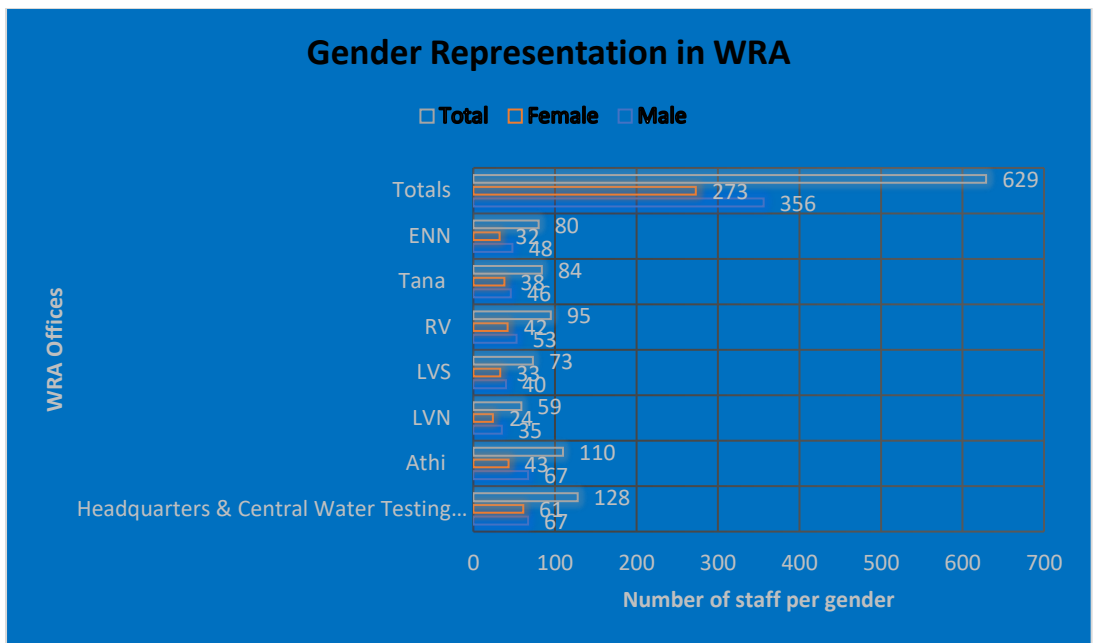


Figure 31: WRA Gender Representation

3.8 Complaint's resolution

The Commission on Administrative Justice (CAJ) is established pursuant to the provisions of Article 59(4) of the Constitution of Kenya, 2010 and the Commission on Administrative Justice Act, 2011. In undertaking its mandate, the CAJ is mandated to investigate complaints against public bodies. To this end, the Authority received various complaints from members of the public. These were resolved and reports on status of resolution submitted to the Commission; 1 complaint was submitted to the Authority from CAJ, a total of 94 complaints were handled directly and 71 complaints were resolved. these are summarized as follows:

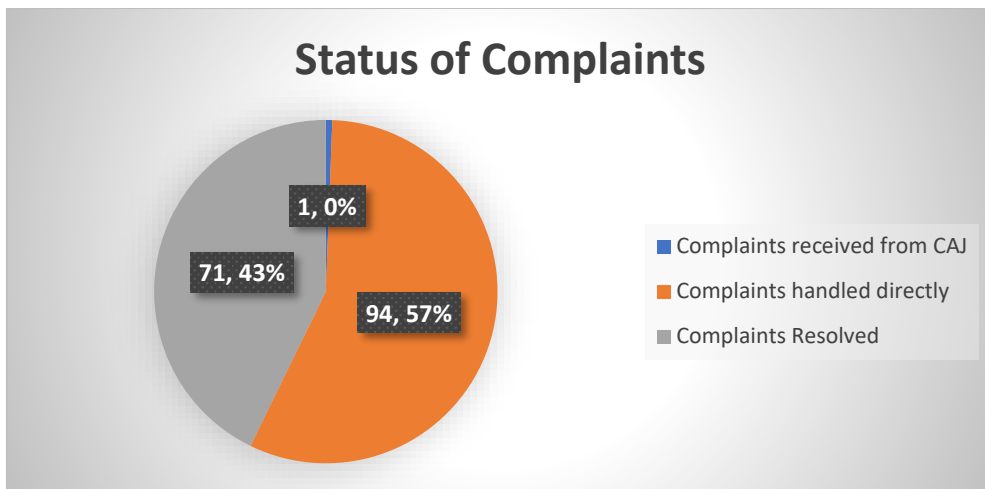


Figure 32: Complaint Resolution

4 Publications

1. Detection, Characterization, and analysis of land subsidence in Nairobi using InSAR Pius Kirui, Samson Oiro, Hunja Waithaka, Patroba Odera, Bjorn Riediel, Markus Gerke
2. Occurrence and environmental risks of contaminants of emerging concerns across the River Athi Basin, Kenya in dry and wet seasons. Faith Chebii, Kenneth K'Koreje, Maurice Okoth, Samuel Lutta , Philip Masime, Kristof Demeestere.
3. Credit Risks Ascendancy on Water-Sanitation Infrastructure Investments in Kenya. *Kabarak Journal of Research & Innovation*, 11(3), 258–269. <https://doi.org/10.58216/kjri.v11i3.70> Dr. Rao Jonnah Owen
Operational Risks Ascendancy on Water-Sanitation Infrastructure Investments in Kenya. *Kabarak Journal of Research & Innovation*, 11(2), 124–134. <https://doi.org/10.58216/kjri.v11i2.80>

Owen, R. J.(2021). Combined influence of operational, credit, liquidity and environmental risks on water-sanitation infrastructure investments in Kenya. *The University Journal*, 3(2), 113-136.

4.https://www.researchgate.net/publication/374387511_The_Stochastic_Hydrological_Forecast_Models_for_Rainfall_Predictions_to_Aid_Climate_Change_Resilience_Within_the_Tana_Alluvial_Aquifer_Zones_Using_the_ARIMA_Box_Jenkins_Algorithms. Meshack O. Amimo PhD

5 Conclusion

The Authority endeavors to serve the public effectively and efficiently in the implementation of its mandate in regulating the management and use of water resources. The automation of key processes in the Authority and onboarding services to E citizen has demonstrated increased access to services and better allocation and use of water resources in Kenya.

Table 17: WRA Basin and Sub Basin Area Contacts

S/No	WRA Basin Area Office	WRA Sub Basin Area Office
1.	Lake Victoria South Basin Area Regional Office P.O Box 666 Kisumu	Northern Shoreline--_ Nyando P.O Box 666 Kisumu
		Southern Shoreline_ Gucha Migori P.O Box 4114 Kisii
		Mara Sondu P. O Box 563 Kericho
2.	Lake Victoria North Basin Area Elgon_Churangányi P. O Box 774 Kakamega	Elgon_Churangányi P. O Box 2764-30200 Kitale
		Kipkaren_Upper Yala P. O Box 3040-30100

		Eldoret
		Lower Nzoia Yala P.O Box 374-40600 Siaya
3	Rift Valley Basin Area P. O Box 1600 Nakuru	Lower_Turkwel P. O Box 73 Lodwar
		Upper Turkwel P. O Box 49 Kapenguria
		Lakes Naivasha-Nakuru P. O Box 66 Naivasha
		South Rift Valley P.O Box 1029 Narok

		Lakes Baringo- Bogoria P. O Box 544-30400 Kabarnet
3	TANA P.O Box 1930-60100 Embu	Upper Tana P.O Box 460-10200 Murang'a
		Tiva/Tyaa P.O Box 781-90200 Kitui
		Thiba P.O Box 1291-10300 Kerugoya
		Kathita/Mutonga P.O Box 3256-60200 Meru

		Lower Tana P.O Box 31-70100 Garissa
5	ATHI P.O Box 1159-90100 Machakos	Nairobi P.O Box 18150-00500 Nairobi
		Middle Athi P. O Box 176-90137 Kibwezi
		NolTuresh Lumi P.O Box 53-00209 Loitoktok
		Upper Athi P.O Box 1864-00900 Kiambu

		Coastal Athi P.O Box 80100-85672 Mombasa
6	Ewaso Ng'iro North Basin Area P.O Box 1331-10400 Nanyuki	Engare Narok Melghis P.O Box 631-20300 Nyahururu
		Ewaso Daua Laggas P.O Box 308-70300 Mandera
		Upper Ewaso Ng'iro P. O Box 1331-10400 Nanyuki
		Middle Ewaso Ng'iro P. O Box 58-60300
		North Ewaso Laggas P.O Box 207-60500

Report Coordinated and Compiled by **Elizabeth Diego**

Manager Resource Mobilization and Stakeholder Engagement

Water Resources Authority

1.0 INTRODUCTION

The Government has used Performance Contracting as a key accountability tool since 2003 in its endeavor to improve service delivery in the Public Service. Performance Contracting is part of the broader public sector reforms that was aimed at improving efficiency and effectiveness in the management of the Public Service.

Taking cognizance of the emerging realities of COVID-19, the Ministry of Public Service and Gender developed a complete end-to-end online platform –the Government Performance Contracting Information System (GPCIS) towards automating the Performance Contracting process. The system has been used for the last 4 years and has eased the Performance Contracting Process.

The Authority through the Board of Management signed its Performance Contract with the Ministry of Water, Sanitation & Irrigation and The National Treasury affirming its commitment to undertake the set targets.

2.0 CASCADING OF THE PERFORMANCE CONTRACT

The 19th Cycle Performance contracting guidelines used to prepare the FY 2022/23 WRA PC clearly indicates; on the Implementation of Performance Contracts, the signed PC should be cascaded by signing lower-level contracts with departments and downstream institutions. In our case, this was adhered to and all Departments/Divisions and Basin areas signed their Cascaded PCs with the Management. This in turn ensured that the Implementation was done at all levels

3.0 IMPLEMENTATION OF THE PERFORMANCE CONTRACT

In Implementation of the PC, The Medium-Term Plan III, Big 4 Agenda, Sustainable Development Goals and the Strategic Plan 2018-2022 were all taken into cognizance. Keen to us was the Rapid Results Initiatives (4th & 5th Waves) which ensured excellence on the Core Mandate Indicators.

Quarterly reports were prepared, discussed and approved by the board of management as required by the 19th Cycle PC Guidelines. These reports were submitted to the line Ministry, The inspectorate of State Corporations, Public Service Performance Management & Monitoring Unit (through the system) and The National Treasury & Economic Planning.

4.0 PERFORMANCE EVALUATION

The Annual Performance Evaluation for the year under review was conducted by the Public Service Performance Management Unit where the Authority scored **2.0473** Translating to *“Excellent”*.

Upper	Lower	Rating
1.00	2.40	Excellent
2.40	3.00	Very Good
3.00	3.60	Good
3.60	4.00	Fair

Key: PC Scoring Criteria

N/B. The annual report indicating the performance of Public Institutions is yet to be released.

5.0 HIGHLIGHT OF ACHIEVEMENTS

The following is a highlight of achievements:

A. FINANCIAL STEWARDSHIP & FISCAL DISCIPLINE

The total amount of funds received in the financial year 2022/2023 was 1,530,750,000 i.e. Kshs. 1,304,000,000 for Re-current Budget, including A-in-A and Kshs. 226,750,000 for Development Budget. The expenditure in A-I-A includes salaries which were partially supported by recurrent grant. Absorption of Recurrent & Development funds were at 100%, generally the absorption was good.

A in A the Authority had a target of Ksh 600,000,000 but we were able to collect Kshs. 870,702,299.63, surpassing its annual PC target by 270,702,299.63. The reason for the over achievement was due to collection of long outstanding debt held by water service providers.

Eternally mobilized funds received and utilized during the FY 2022/23 amounted to Kshs. 110,424,091.20 i.e. Kshs. 57,039,041.20 from KWSCR-World Bank and Kshs. 53,385,050.00 from Towards Ending Drought Emergencies (TWEDE) Project.

As at the end of FY 2022/23, the Authority had cleared all its pending bills incurred within the FY.

B. SERVICE DELIVERY

The Authority excelled in service delivery by undertaking the following;

- ✓ All WRA stations prominently displayed the Service Delivery Charter in the correct format, three feet in width and four feet in height at the points of entry. In addition, the Authority customized the service delivery charter to braille and sign language, sensitized 125 members of staff on customer excellence and analyzed feedback from customers on service delivery.
- ✓ The Authority re-engineered the following 3 processes; a) Revenue management by converting paper-based receipts to automated system receipts. b) Meter reading by adopting a mobile App that helps in capturing the meter reading and transmission of the data into the water meter data management system in real-time. c) Billing by dispatching payment invoices to customers through SMS and e-mail.

- ✓ During the FY under review, the Authority performed as follows with regard to complaint resolutions: Complaints received from CAJ - 1, Complaints handled directly-94, Actual resolution of complaints - 71. The Authority also submitted all quarters' performance reports to CAJ on time.

The Authority was issued with certificates/scores on service delivery as follows:

	Indicator	Specialized Agency	Score %
B1	Citizens Service Delivery Charter	Public Service Performance Management Unit (PSPMU)	100
B2	Business Process re-engineering	Public Service Transformation Department	100
B3	Resolution of Public Complaints	Commission on Administrative Justice (CAJ)	97

C. CORE MANDATE

The Authority made numerous achievements with regard to its core mandate in the FY 2022/23 as follows:

C1. Water Resources Monitoring Stations Constructed and Rehabilitated

- ✓ The following achievements were made in the FY under review: a) Rehabilitated 24 Regular Gauging Stations (RGS) b) Established 2

rainfall stations c) Upgraded 1 station (3BA29) to telemetry d) Maintained operations of 116 telemetric stations.

C2. Exploratory Boreholes Drilled

- ✓ The Authority carried out hydrogeological assessments for deep aquifers for 34 sites.
- ✓ Drilling of 13 boreholes was carried out as follows: 7 in Tiwi, 1 manguo, 1 Nchura, 1 Narok. 1 Manguo and 2 boreholes in Kakibei.
- ✓ Hydrogeological profiling of 3 Aquifers was done as follows: Nairobi aquifer suite, Mt. Elgon and Kakamega. This was done through analysis of borehole completion records, the use of borehole cameras and geophysical logging.

C3. Sub-Catchment Management Plans Implemented

- ✓ The Authority operationalized and capacity-built KeNAWRUA by establishing and furnishing a physical office, conducting a two-day induction of its members and sensitizing members on Water Resources Regulations 2021.
- ✓ The Authority formed 2 WRUAs in Pate and Tangulbei Sub-catchments in Lower Tana (Garissa) and Kabarnet sub-basin areas, respectively.
- ✓ The Authority constructed 3 sand dams as follows: i) Kalundu WRUA in Kitui County constructed 2 sand dams, namely Kwa Mweu-Kalimenza and Kalimangovu sand dams under the EU EDE CPIRA program. Ilmunkush WRUA in Kajiado County constructed Lesoit sand dam.
- ✓ The Authority and its partners developed 5 SCMPs for Ikoo, Pate, Tangulbei, Middle Kuywa, Nambale and Ikonzo WRUAs and reviewed 1 SCMP for Upper Kuywa WRUA.
- ✓ The Authority installed 8 Rain Water Harvesting Tanks in various institutions through Bukhungu, Mumias and Safu WRUAs.

C4. Resource Mobilization Proposals Developed

- ✓ The Authority developed 16 proposals, namely;
 1. Nature-Based Solutions for Water Resources Management (ABA)
 2. Water Resources Development and Conservation Project (ABA)
 3. Enhancing Access to Water Availability through Water Storage (LVSBA)
 4. Rehabilitation of Critical Water Ecosystems (LVNBA)
 5. Assessment of Olbanita Aquifer (RVBA)

6. Enhancing livelihoods for riparian communities (LVSBA)
7. Rehabilitation of Kwa Kilonzo dam (ABA)
8. Soil and water conservation (LVNBA)
9. Protection and Restoration of Wetlands in (LVNBA)
10. Capacity Development for Integrated Flood Management in Tana Basin (TBA)
11. Water Resources Management in (RVBA)
12. Restoration of Critical Water Ecosystems in (LVNBA)
13. Community-led Water Harvesting and Storage Initiative (ABA)
14. Nzoia Watershed Management (LVNBA)
15. Achieving Gender Equality through Water in Kenya (ALL BASINS)
16. Lake Kenyatta Ecosystem Restoration (TBA).

C5. Water Quality Monitoring

- ✓ The water quality assurance in mid-Tana, Sabarwawa and Chyulu Hills landscapes was undertaken and completion reports were prepared. In addition, water quality assurance was also carried out in Thwake dam and in the six basin areas.
- ✓ The Authority carried out quarterly water quality monitoring and Pollution inspection/ enforcements in all its 6 basin areas as follows: ENNBA (102No-SW, GW & Effluent), LVNBA (139No-SW, GW & Effluent), RVBA (78No. SW, GW & Effluent), ABA (159No-SW ,GW & Effluent), LVSBA (106No-SW, GW & Effluent)

C6. Water Catchment Areas Protected

- ✓ The Authority developed catchment management guidelines for Lari swamp, Karai wetlands, Mwangea hills and Lake Ol'bolossat and prepared draft conservation area orders for each. The four drafted conservation area orders and explanation memoranda were submitted to the attorney general for gazettelement.

C7. National Geo-reference Water Resources Monitoring Information System

- ✓ The Authority conducted a baseline survey in quarter 1, which showed that 22 stations were transmitting data as follows: 16No SW and 6No. GW. In the subsequent quarters, the stations transmitting data were increased by 130 as follows: 63 Surface water, 12 Groundwater quality and 55 groundwater.

C8. Advanced Water Meter Data Management System

- ✓ E-Permit System was successfully integrated with M-Pesa payments and Bank payments on all permitting services. All payment validations are automated and are done by the system. In addition, the billing module was fully developed and in use. Invoices are generated by the system and are dispatched both via SMS and E-mail transmission to the clients.

C9. Water Resources Data Management Strengthened

- ✓ A baseline Survey on the existing hydrological network for Lake Victoria Basin was conducted and a report prepared.
- ✓ Capacity building of WRA staff on hydrological monitoring, assessment and data analysis was conducted.

C10. Project Completion Rate

- ✓ The Water Resources Performance Report for the FY 2021/2022 (the identified project) was prepared and disseminated to the relevant stakeholders.

C11. Water Abstraction & Pollution Surveys Conducted

- ✓ Mapped and geo-referenced of 268 monitoring stations.
- ✓ Implementation of abstraction and pollution recommendations from (Mwania Mbogo, Muringato and Moiben) water abstraction and pollution surveys was done.
- ✓ Carried out 2 water abstraction and pollution surveys (Kiu and Suguroi) ecosystems and reports prepared.
- ✓ Prepared the Water Allocation Plans for Upper Thika Sub basin.

C12. Kikuyu Springs Groundwater Conservation Area Secured and Protected

- ✓ Karai and Lari swamps management guidelines were developed and forwarded to the Ministry of Water, Sanitation and irrigation for gazettelement.
- ✓ Marked and pegged Karai and Lari swamps.
- ✓ Implementation of the Kikuyu Springs groundwater conservation area order was undertaken by Planting of 3,000 trees by Onkaru WRUA, the established management committee of the Kikuyu Springs GCA, installation of 2 rainwater harvesting tanks in Lusigetti Primary school and establishment of 1 rainfall station in Murengeti Primary school.

C13. Athi River Restoration Programme

- ✓ Calibration of 27 Instruments/Equipment in Embu Basin Laboratory and Central Water Testing Lab in Nairobi.

- ✓ The Authority collected and analyzed 188 data sets from stations upstream of Thwake dam.
- ✓ Dissemination of the developed guidelines for solid waste management upstream of Thwake Dam to stakeholders.

C14. Water Abstracted Under Permit Enforced

- ✓ During the FY 2022/23, the volume of water abstracted under permit was increased by 2,306,020.54 m³/day against a target of 100,000 m³/day.

C15. Inspections of Expired Authorizations and Permits Undertaken

- ✓ During the FY under review, 1018 authorizations were converted into new permits, while 100 expired permits were renewed.

C16. Lamu Groundwater Conservation Area Secured and Protected

- ✓ Training on groundwater modeling carried out for 30 No. Officers.
- ✓ Implementation of the Lamu groundwater conservation area order was carried out by constructing 1 Djabia in lake Kenyatta by Amu WRUA, planting 10,000 trees in the sand dunes by Amu WRUA, Establishing the Management Committee of the Amu GCA, Constructing and equipping 7 No. monitoring wells in the sand dunes, and installation of 1 rainfall station in the compound of LAWASCO.
- ✓ Carried out stakeholders' consultative workshop for Lamu Water Allocation Plan.

C17. Rating Equations Updated

- ✓ Long-Term flow measurements were compiled for a total of 40 RGSs completed.
- ✓ Cross-Sectional Survey for 18 stations for updating of Rating Curves has been initiated.
- ✓ 12 Rating Curves updated and approved for the following stations. 1GD03 Nyando, 1AH01 Sio River, 1KB05 Kuja Migori River, 1FG02 Yala River, 2EE07B Perkerra River, 2B33 Suam River, 3DA02 Athi River at Munyu, 3GA02 Tsavo River, 4BE01 Maragua, 4G01 Tana River at Garissa, 5BE01 Nanyuki and 5ED01 Ewaso Ng'iro River.

C18. Dam Safety

- ✓ Standards, Guidelines and Procedures for water structures and Dam Safety were formulated and are in place.
- ✓ Codes of Practice for the Evaluation of Dam Designs, Construction, Supervision and Inspection of Dams have been developed and are in place.
- ✓ In-housing capacity building was done for 15 Water Resources Safety Officers on Dam safety.
- ✓ SD2 and SD3 Dam's databases are established and regularly updated.
- ✓ Sensitization of Dam owners/Operators on Dam Safety.

D. IMPLEMENTATION OF PRESIDENTIAL DIRECTIVES

In order to fulfill the Presidential Directives issued, the following activities were undertaken in the FY 2022/23:

- i. Planted 66, 088 tree seedlings in collaboration with stakeholders.
- ii. Wrote a memo to staff informing them to be dressed in decent, smart casual Kenyan-produced clothes on Fridays, which they adhered to.
- iii. Promoted use of cashless transactions in revenue collection (126.893 million was paid through Mpesa)
- iv. Avail hand sanitizers at the office entry points.

E. ACCESS TO GOVERNMENT PROCUREMENT OPPORTUNITIES AND

F. PROMOTION OF LOCAL CONTENT IN PROCUREMENT

The procurement was distributed and achieved as follows:

Procurement Plan Implementation Status FY 2022-2023				
	All Procurement	AGPO 30% (E-1)	PWD 2% (E-iv)	Buy Kenya Build Kenya 40%
Availed procurement Budget in the FY 2022/23	242,896,352.38	73,790,695.00	2,253,850.00	242,896,352.38
Percentage Achieved based on Availed Procurement Budget		30.38%	3.09%	100%

This achievement was attributed to the following:

- Prequalification of suppliers' tender advert was placed and a list was approved subsequently.
- All the Procurement Reports are submitted quarterly to the PPRA Via PPI Portal.
- All procurement opportunities allocated and awarded target groups were paid Promptly.
- All the Procurement Reports allocated to PWDs were submitted quarterly to the NCPWD through the portal www.ncpwd.co.ke
- In addition, all quarterly reports were submitted to Ministry of Industrialization, Trade and Enterprise Development
- The Authority successfully updated the catalogue of assets and properties a tender document for disposal of obsolete, surplus and Unserviceable assorted items.

G. CROSS CUTTING

The following achievements were realized during the year under review:

	Indicator	Specialized Agency	Score %
G1.	Asset Management	The National Treasury & Economic Planning	89.25
G2.	Youth Internship/Apprenticeships/ Attachments	As Evaluated by the PSPMMU	301
G3.	Competence Development	As Evaluated by the PSPMMU	100
G4	Disability Mainstreaming	As Evaluated by the PSPMMU	100
G5	Gender Mainstreaming	State Department for Gender/ National Gender & Equality Commission (NGEC	97
G6	Prevention of Alcohol and Drug Abuse	National Council for Alcohol & Drug Abuse (NACADA)	92.50
G7	Prevention of HIV Infections and Non-Communicable Diseases	National AIDS Control Council (NACC)	100
G8	National Cohesion and Values	Directorate of National Cohesion & Values	100

G9	Road Safety Mainstreaming	National Transport & Safety Authority (NTSA)	100
G10	Corruption Prevention	Ethics & Anti-Corruption Commission (EACC)	97

CONCLUSION:

The Performance Contract Overall Implementation was Excellent.



Securing water resources, safeguarding our future

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